A unified description of low-energy fusion and multinucleon-transfer reactions within the Coupled Master-Langevin model

Speaker：Zehong Liao

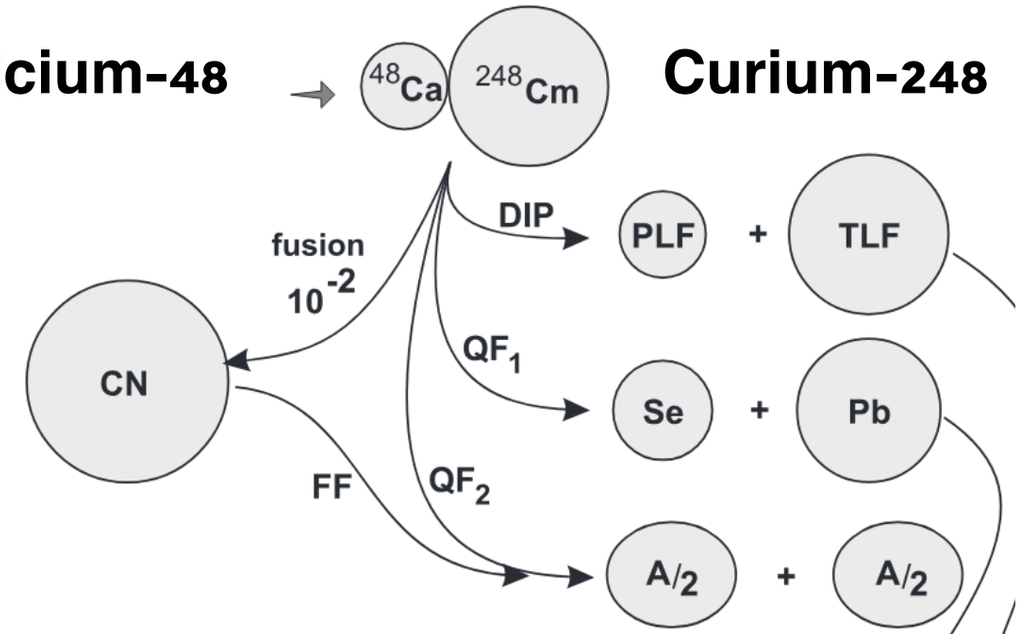
Content

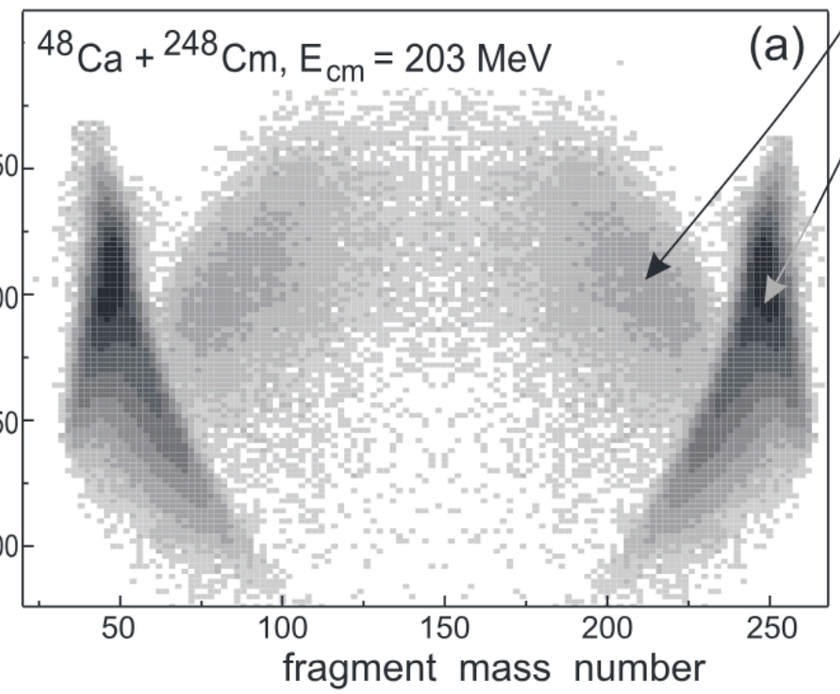
A. Background
B. Degree of freedom
C. Potential energy surface
D. Equations of motion and nucleon transfer
E. Modeling nucleus-nucleus collisions
F. Summary

Seminar at Kyoto University, Oct. 16

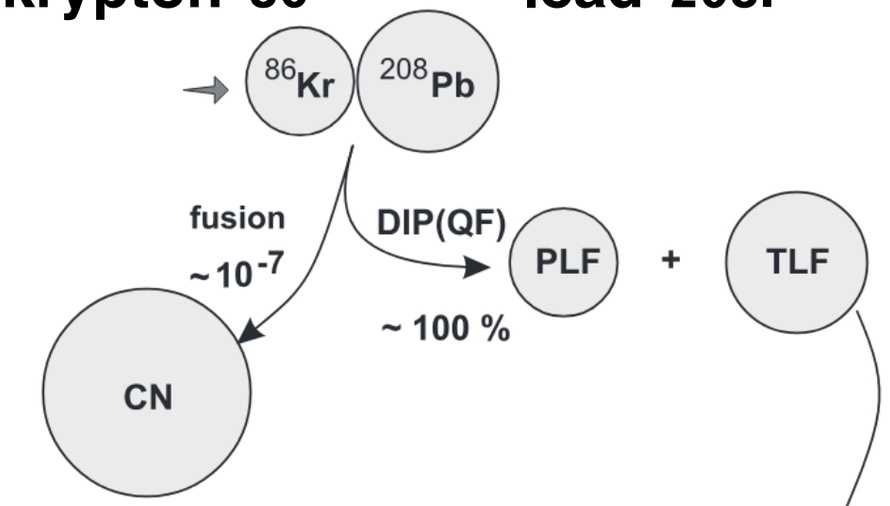
What can we see from the low energy collision, E < 10 MeV/u

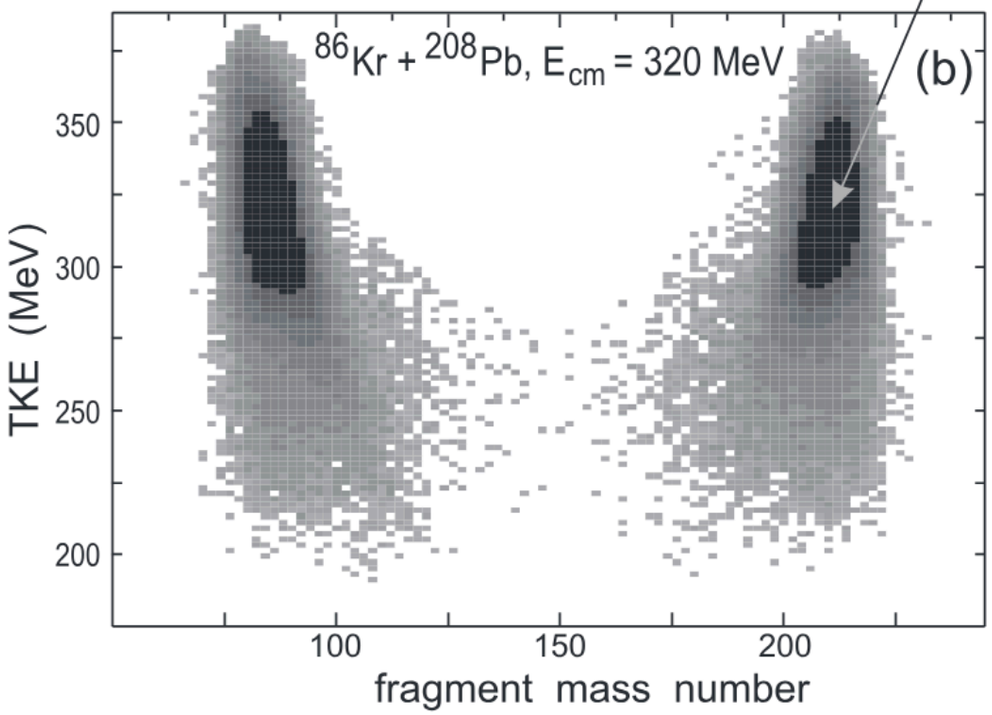
Calcium-48 → ^{48}Ca ^{248}Cm **Curium-248**





krypton-86 → ^{86}Kr ^{208}Pb **lead-208.**





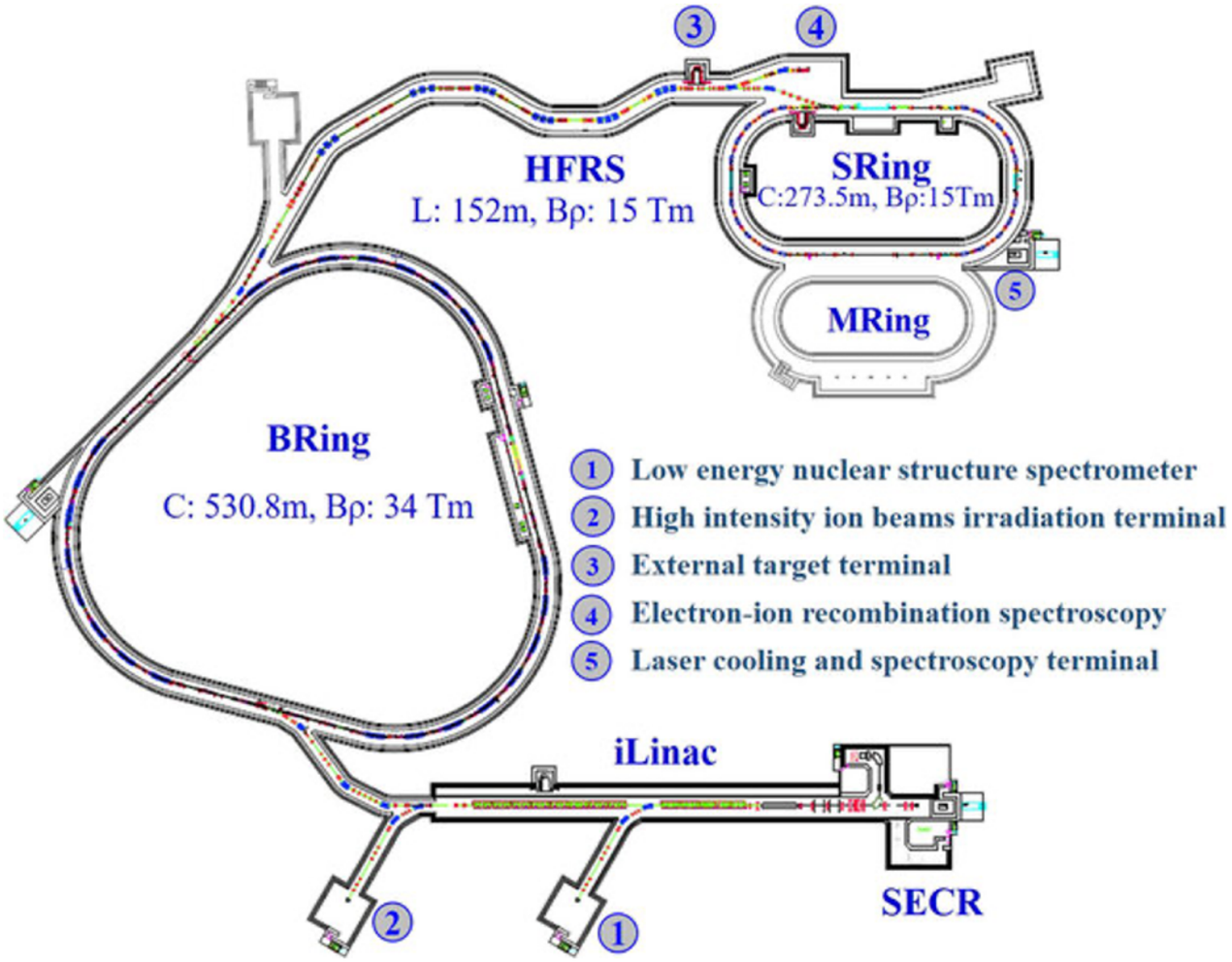
Experiment result: the correlation between the fragment mass and fragment energy

DIP: Deep inelastic process

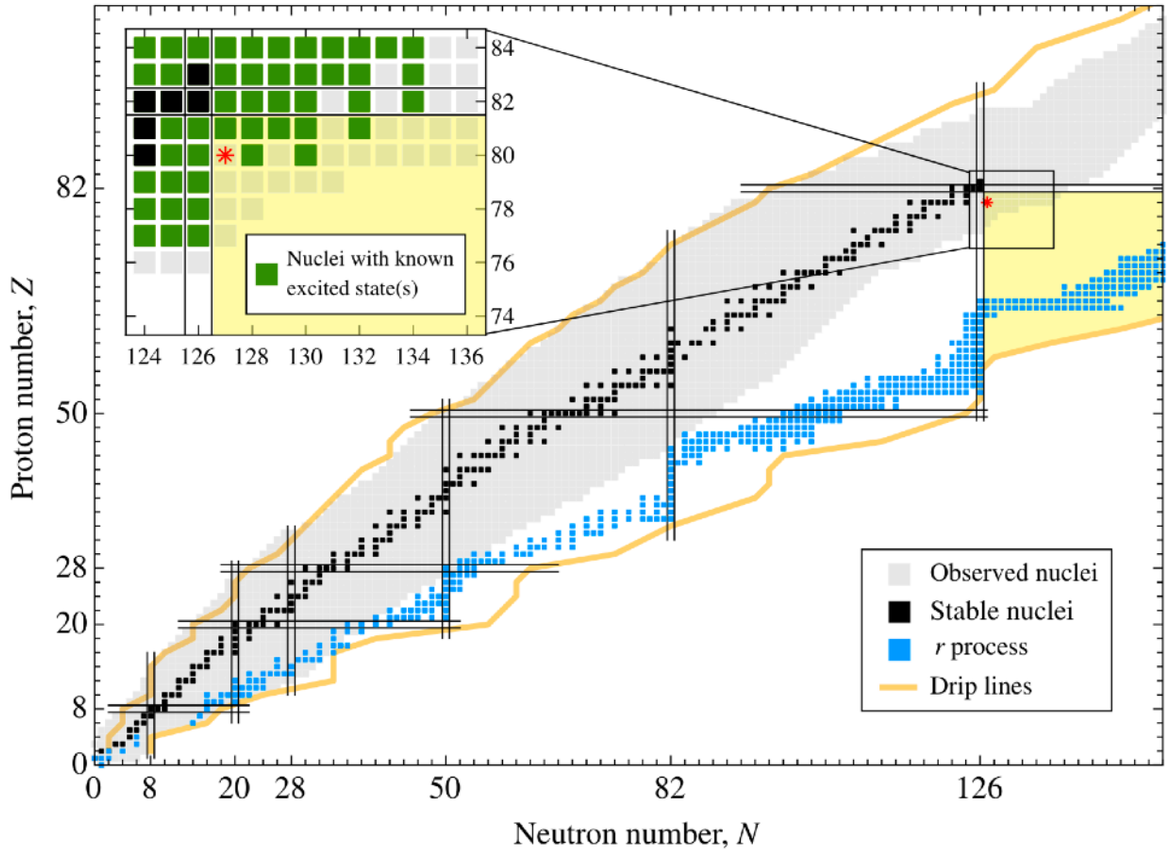
QF: quasi-fission

Itkis M G et al 2001 Proc. Int. Conf. on Fusion Dynamics at the Extremes (Dubna) ed Yu Ts Oganessian and V I Zagrebaev (Singapore: World Scientific) p 93

Experiment terminal in HIAF, China



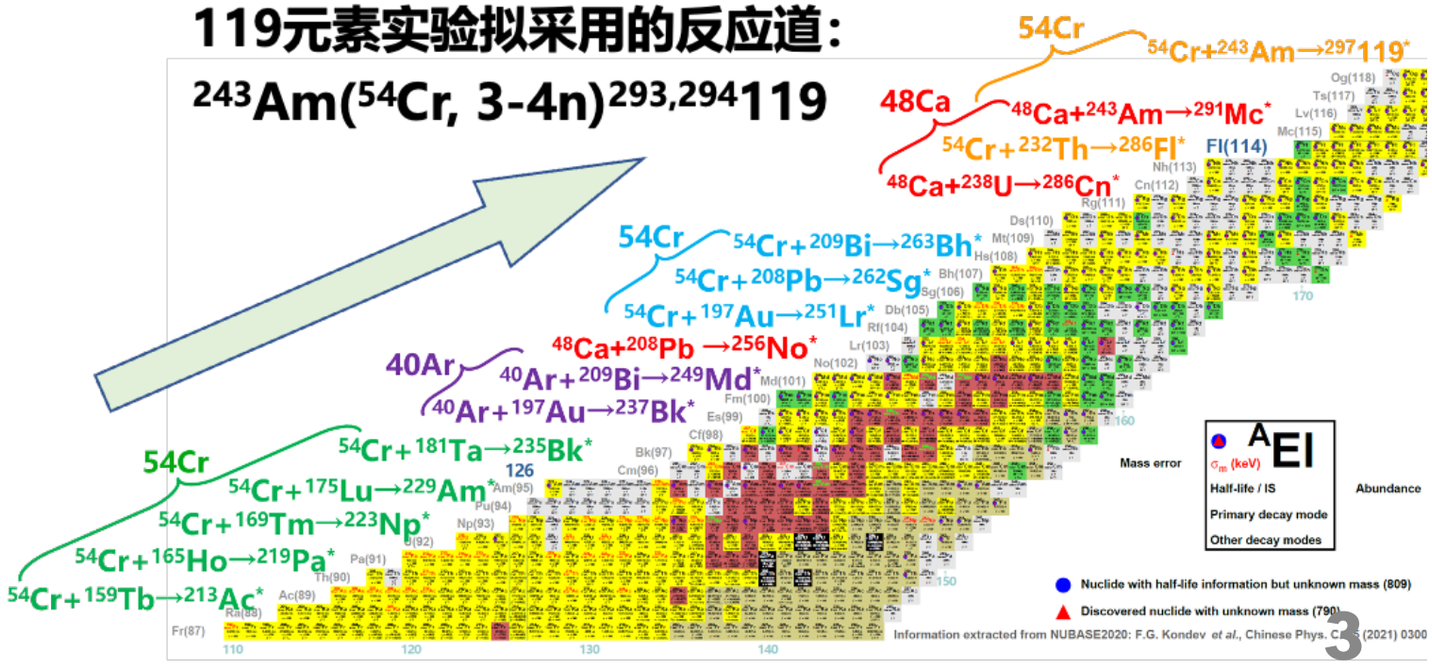
New isotones at N =126



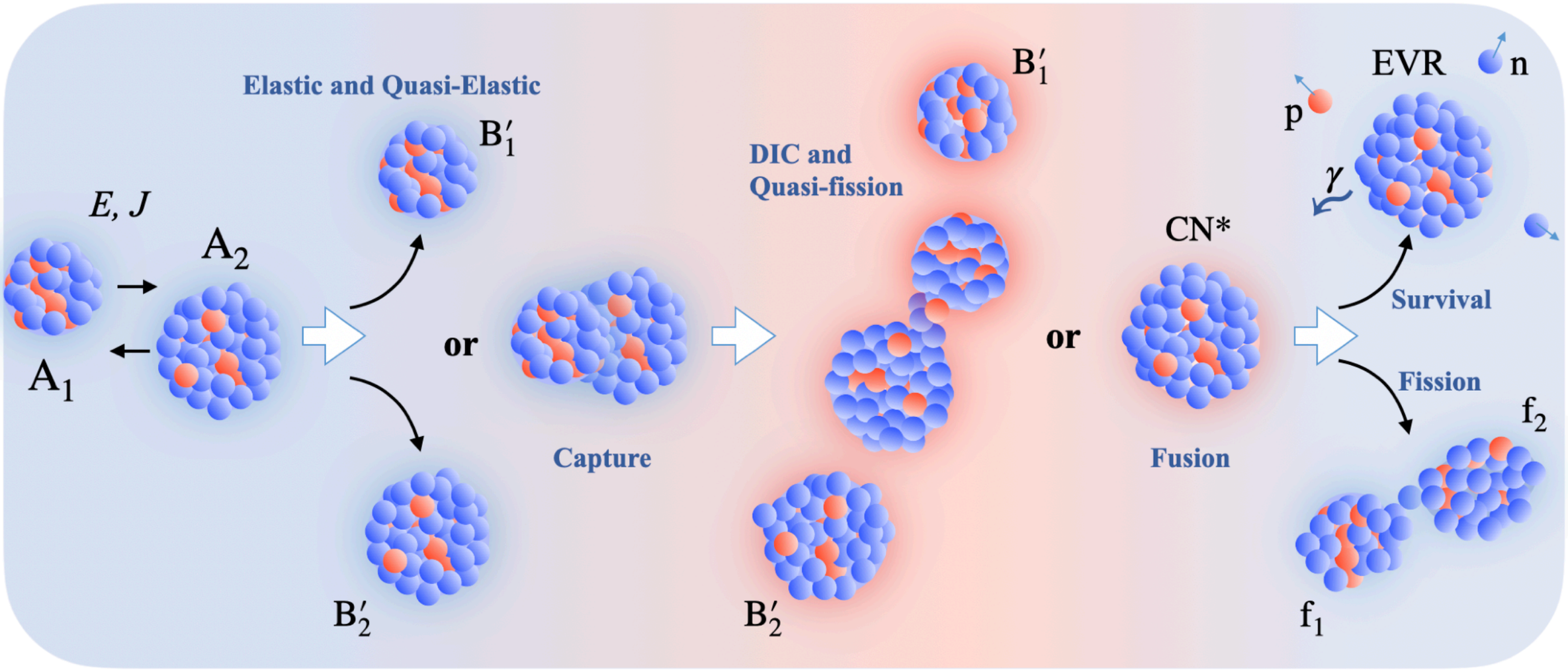
New Superheavy element

119元素实验拟采用的反应道：

$^{243}Am(^{54}Cr, 3\text{-}4n)^{293,294}119$



Modeling nucleus-nucleus collisions



$$\sigma\left(Z_f, N_f\right) = \frac{\pi\hbar^2}{2\mu E_{\text{c.m.}}} \sum_{J=0}^{J_{\max}} (2J+1) P_{\text{cap}}(J) \sum_{Z_1, N_1} P(Z_1, N_1; J^*, E^*) W(Z_1, N_1 \rightarrow Z_f, N_f; J^*, E^*),$$

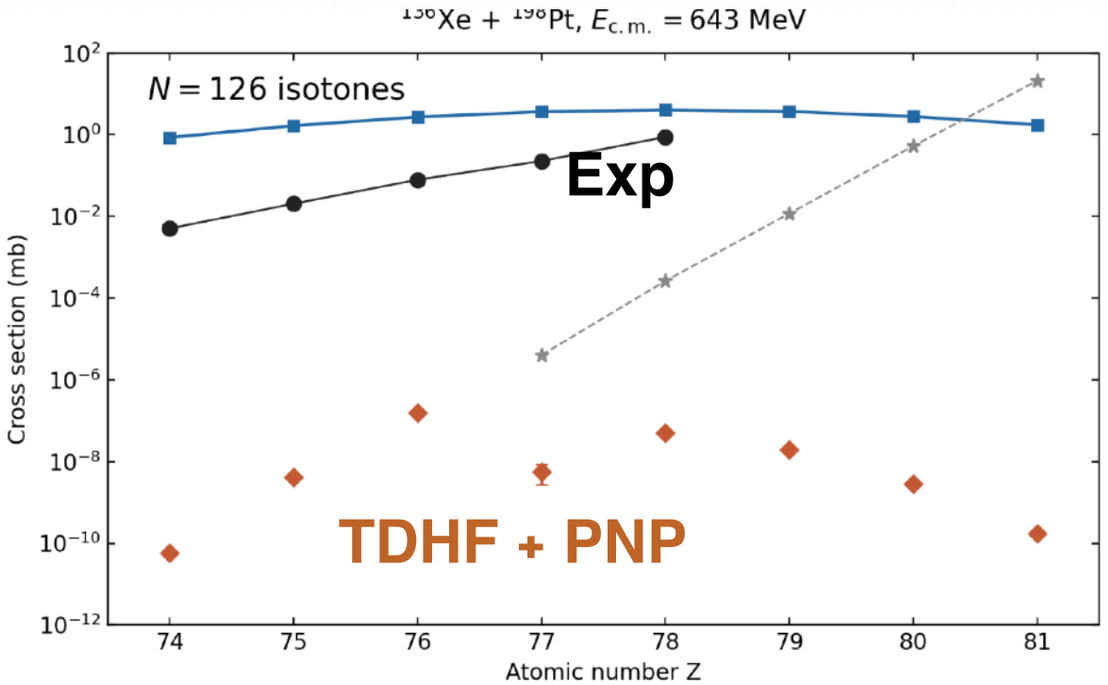
in generally, this term is the aspect that theorists most want to determine, yet it is also the most difficult to determine.

Different angle to the low energy collision

Microscopic — Nucleonic Degrees of Freedom

Starting point:	Self-consistent evolution of indivi
Method:	TDHF
Strength:	Microscopic connection between
Limitation:	Mean-field approximation underes

Nucleonic degrees of freedom ⟷ *Co*



Macroscopic — Collective Degrees of Freedom

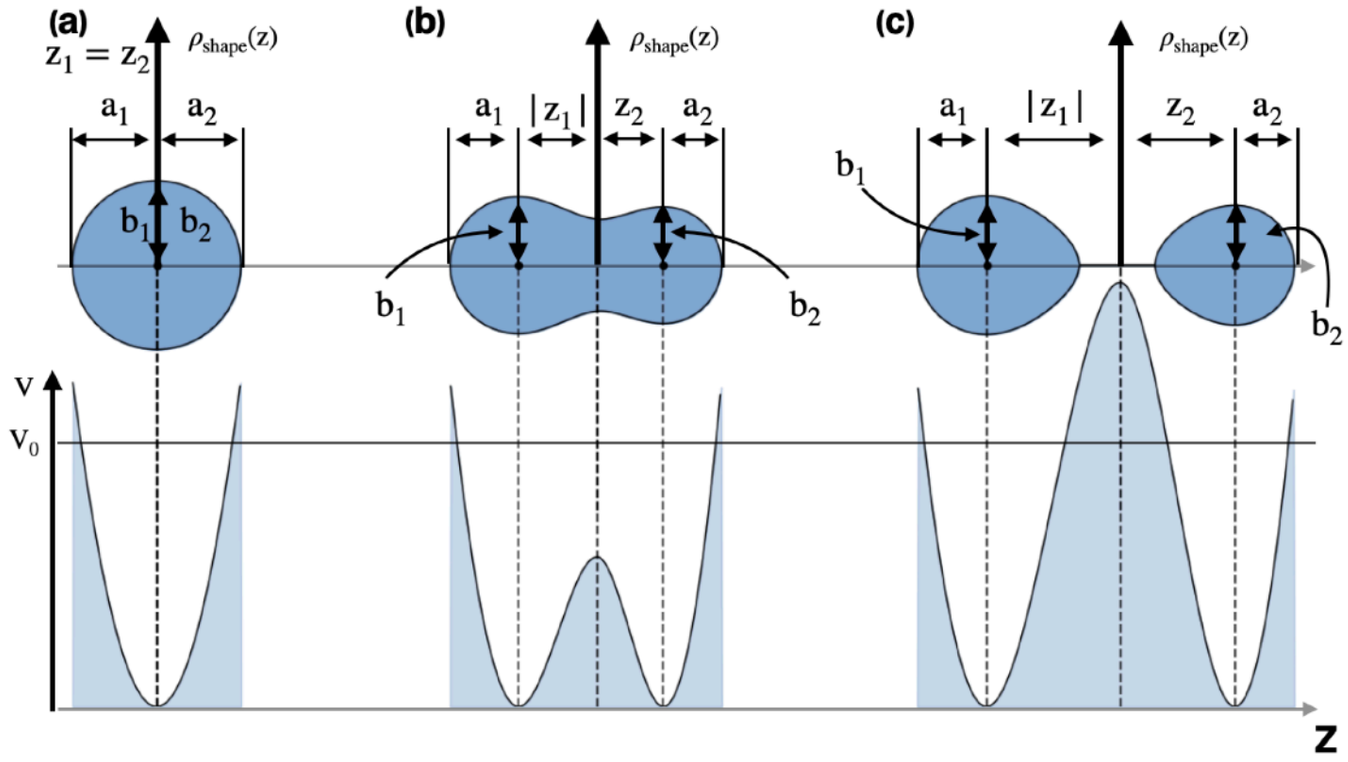
Starting point:	Evolution of a few collective variables, e.g. R, δ, η_A, N, Z
Methods:	DNS / Langevin
Strength:	Efficient description of dissipation, diffusion, and broad fragment distributions
Limitation:	Requires modeled PES and transport/transfer coefficients

Degree of freedom - static degree of freedom

The parametrization originating from the two center shell model (TCSM)

$$\rho^2_{\text{shape}}(z) = \begin{cases} b_1^2\left[1-\dfrac{(z-z_1)^2}{a_1^2}\right], & z \le z_1, \\ b_1^2\left[1-\dfrac{(z-z_1)^2}{a_1^2}(1+c_1z'+d_1z'^2)\right]\dfrac{1}{(1+g_1z'^2)}, & z_1 < z < 0, \\ b_2^2\left[1-\dfrac{(z-z_2)^2}{a_2^2}(1+c_2z'+d_2z'^2)\right]\dfrac{1}{(1+g_2z'^2)}, & 0 < z < z_2, \\ b_2^2\left[1-\dfrac{(z-z_2)^2}{a_2^2}\right], & z \ge z_2, \end{cases}$$

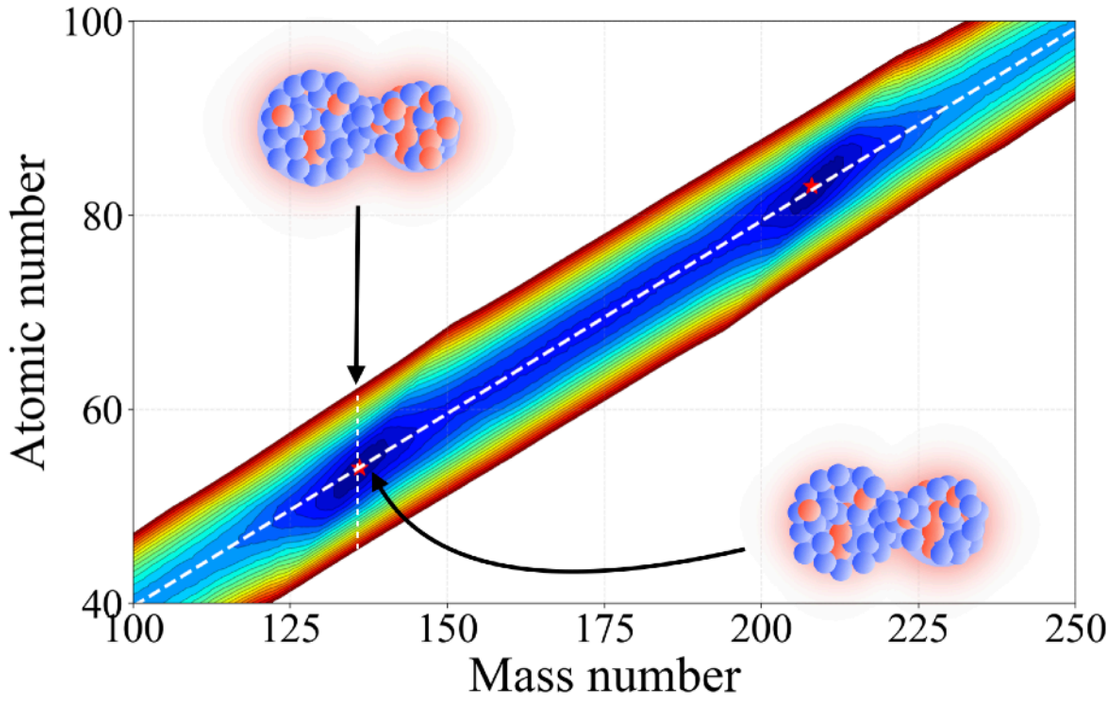
$$z' = \begin{cases} z-z_1, & z \le 0, \\ z-z_2, & z > 0. \end{cases}$$

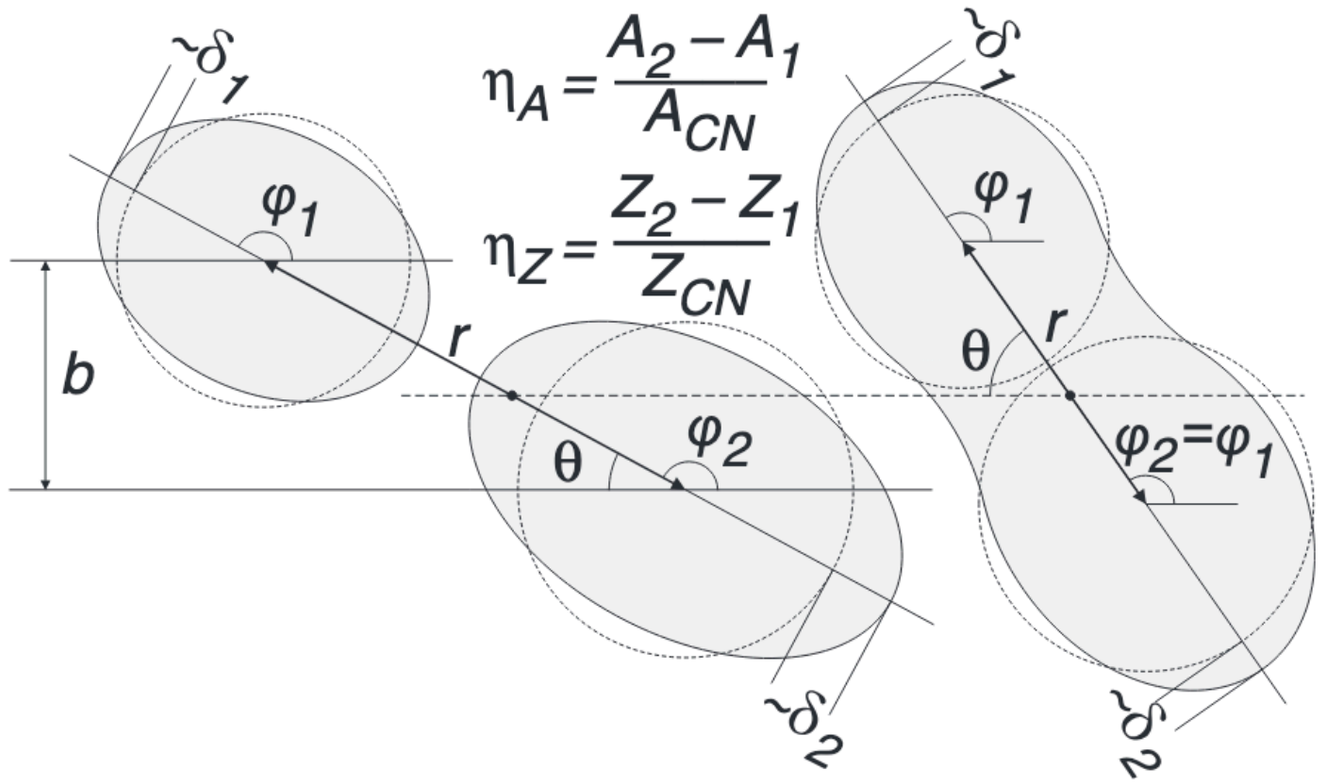


12 shape parameter reduced to five Degree of freedom

- $r = z_1 - z_2$: **The distance between the oscillator centers,**
- $\delta_{1(2)}$: **The Ellipsoidal deformations, defined as** $a_i/b_i - 1$
- η_A: **The mass asymmetry, defined as** $A_2 - A_1/(A_1 + A_2)$
- ε: **The neck parameter. The smaller ε is, the thicker the neck**

Degree of freedom - dynamical degree of freedom





- η_Z : the **charge asymmetry** between the left and right side
- θ : the **angle** between the symmetry axis and the beam direction
- $\varphi_{1(2)}$: the **rotation angle** of the left (right) nuclei

Degree of freedom



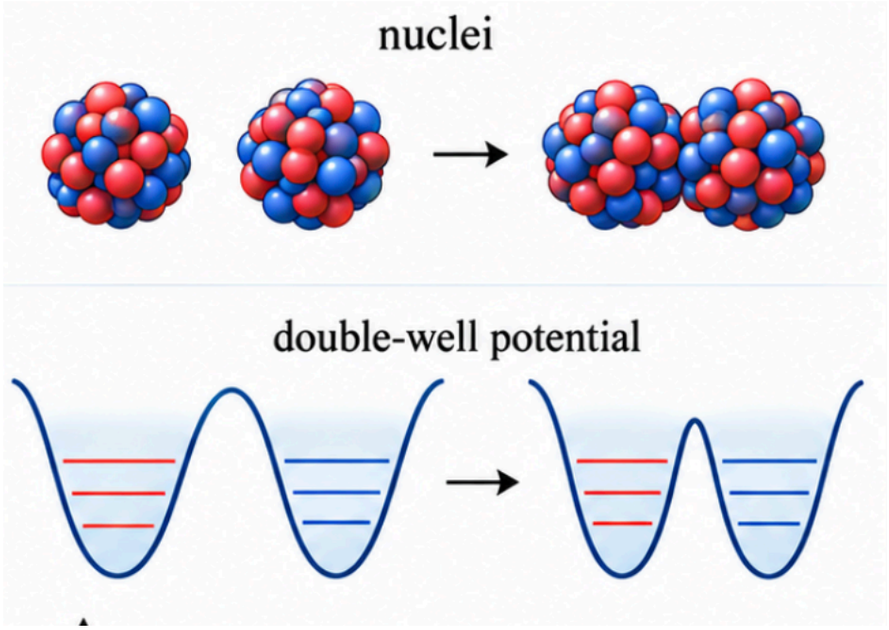
In this model, we consider seven degree of freedom, the collective coordinate include:

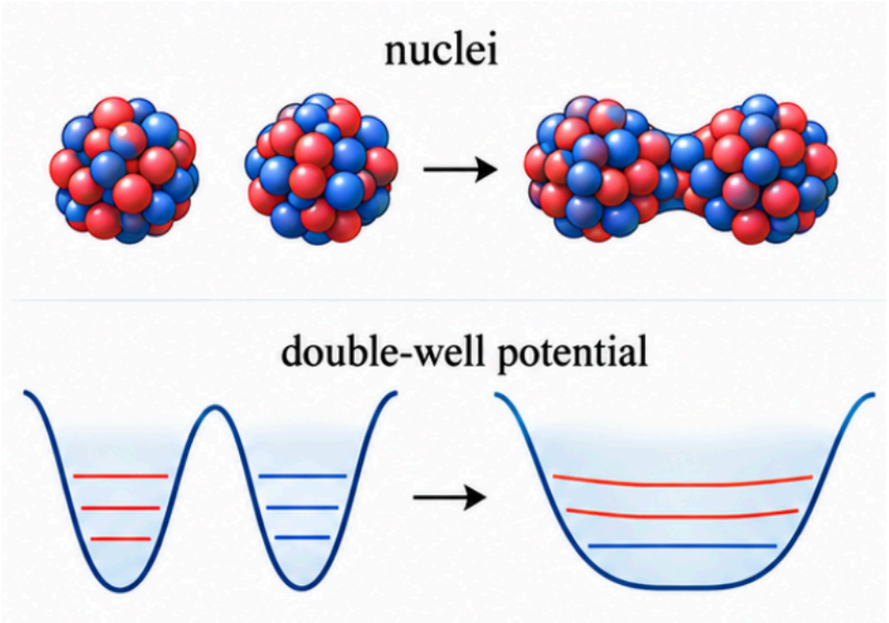
$$q_i = \{r, \delta_1 = \delta_2, \eta_A, \eta_Z, \theta, \varphi_1, \varphi_2\}$$

- r : **the distance between two mass center**
- $\delta_{1(2)}$: **the deformation of the left (right) side**
- η_A : **the mass asymmetry between the left and right side**
- η_Z : **the charge asymmetry between the left and right side**
- θ : **the angle between the symmetry axis and the beam direction**
- $\varphi_{1(2)}$: **the rotation angle of the left (right) nuclei**

V. Zagrebaev, A. Karpov, Y. Aritomo, M. Naumenko, and W. Greiner, Phys. Part. Nucl. 38, 469 (2007).

Potential energy surface





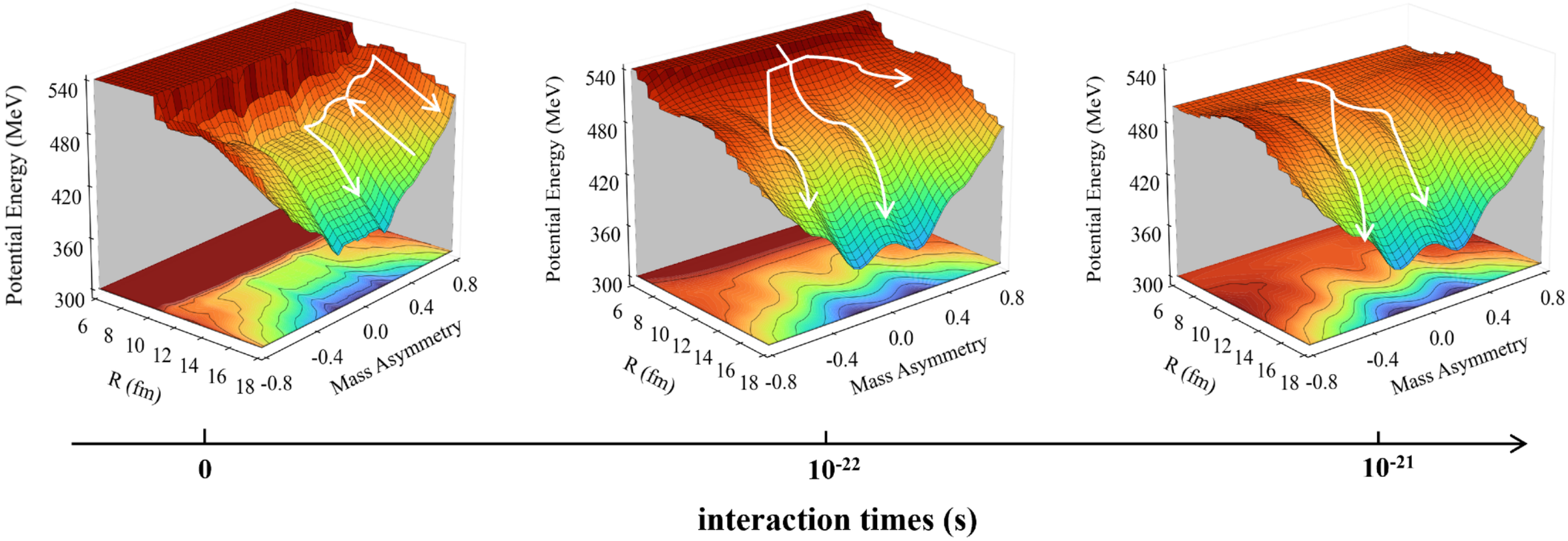
Then, we connect the diabatic and adiabatic potential energies with a time-dependent weighting function $f(t)$ as follows:

$$V(q,t) = V_{\text{diab}}(q)f(t) + V_{\text{adiab}}(q)[1 - f(t)] \qquad f(t) = \exp(-\frac{t}{\tau_{\text{DA}}}), \quad \tau_{\text{DA}} = 10^{-22}\text{s}$$

Where q is a set of collective coordinates used

V. Zagrebaev, A. Karpov, Y. Aritomo, M. Naumenko, and W. Greiner, Phys. Part. Nucl. 38, 469 (2007).

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V. Zagrebaev, A. Karpov, Y. Aritomo, M. Naumenko, and W. Greiner, Phys. Part. Nucl. 38, 469 (2007).

Potential energy surface - Diabatic potential energy

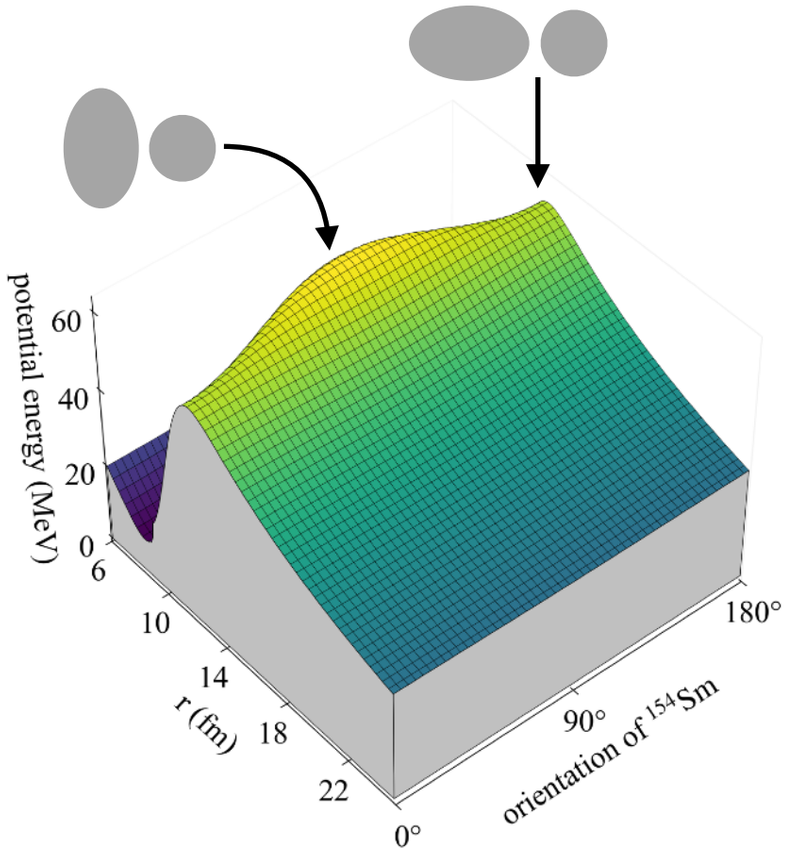
Redistribution of nucleons between separated nuclei changes their binding energies. The diabatic potential energy is defined as follows:

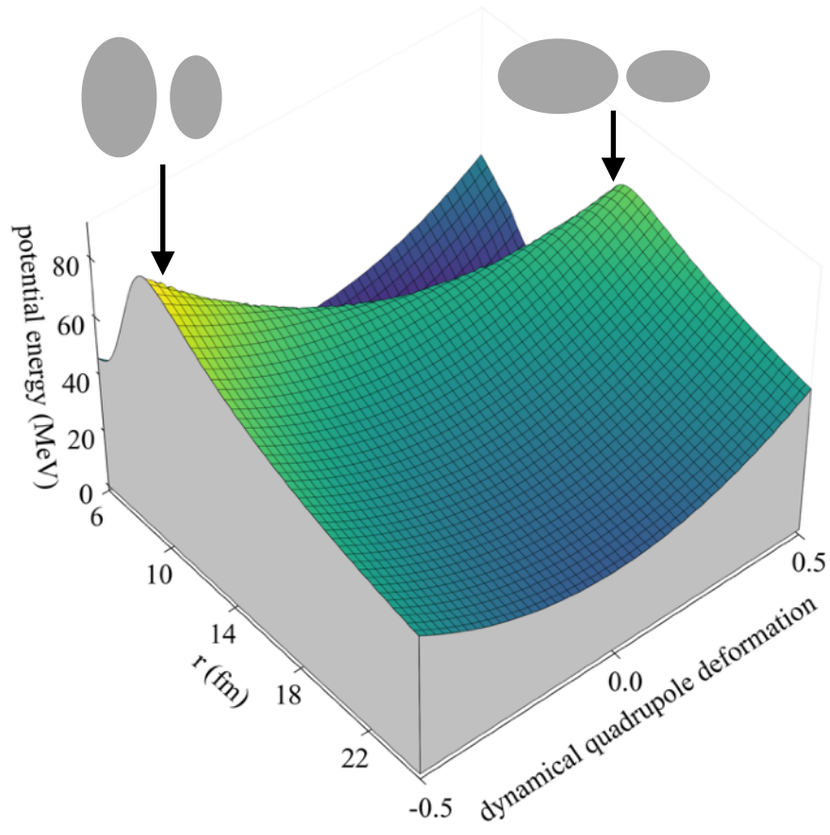
$$V_{\mathrm{diab}}(A,Z;q) = V_{12}(A,Z;q) + M(A_1,Z_1;\delta_1) + M(A_2,Z_2;\delta_2) - M(A_T,Z_T;\delta_T^{\mathrm{g.s.}}) - M(A_P,Z_P;\delta_P^{\mathrm{g.s.}})\,.$$

the masses of the reaction fragments

The interaction energy of the nuclei

the masses of the entrance channel





Potential energy surface - Adiabatic potential energy

The adiabatic potential energy is defined as a difference between the mass of the whole nuclear system and the ground-state masses of the target and the projectile,

$$V_{\rm adiab}(A,Z;q) = M(A,Z;q) - M(A_T,Z_T;\delta_T^{\rm g.s.}) - M(A_P,Z_P;\delta_P^{\rm g.s.})\,.$$

the mass of the whole nuclear system

the masses of the entrance channel

The standard macro-microscopic model is often used for calculation of the total mass:

$$M(A,Z;q) = M_{\rm mac}(A,Z;q) + \delta E(A,Z;q)\,.$$

- $M_{\rm mac}(A,Z;q)$ **is macroscopic liquid-drop mass**
- $\delta E(A,Z;q)$ **is the microscopic shell correction by Strutinsky shell-correction method**

Potential energy surface - Adiabatic potential energy

$$M_{\mathrm{mac}}(A,Z;\vec{q})\,|_{n_Z=n_A} = m_pZ + m_nN$$ ---------- mass excesses of Z hydrogen atoms and N neutrons

$$-a_v(1-k_vI^2)A + a_s(1-k_sI^2)B_N(\vec{q})A^{2/3}$$ ---------- volume energy, surface energy

$$+\frac{3}{5}\frac{e^2Z^2}{r_0A^{1/3}}B_C(\vec{q}) - \frac{3}{4}\frac{e^2}{r_0}\left(\frac{9Z^4}{4\pi^2A}\right)^{1/3}$$ ---------- Coulomb energies, the Coulomb exchange correction

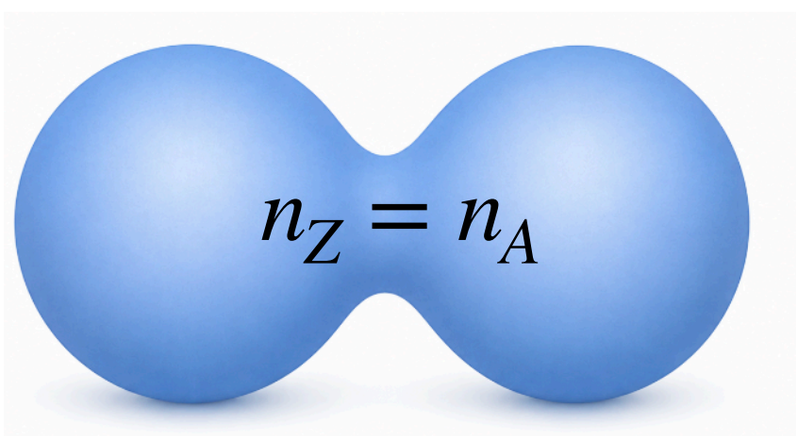
$$+f(k_Fr_p)\frac{Z^2}{A} - c_a(N-Z)$$ ---------- the proton form-factor correction, the charge-asymmetry energy

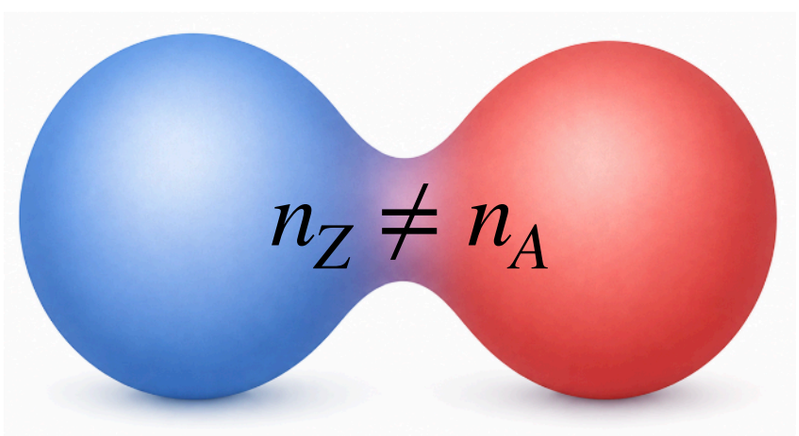
$$+a_0$$ ---------- the constant term

$$+W\left(|I| + \begin{cases} 1/A, Z \text{ and } N \text{ equal and odd} \\ 0, \text{ otherwise} \end{cases}\right)$$ ---------- the Wigner energy

$$+\begin{cases} \Delta_p + \Delta_n - \delta_{np}, Z \text{ and } N \text{ odd} \\ \Delta_p, Z \text{ odd and } N \text{ even} \\ \Delta_n, Z \text{ even and } N \text{ odd} \\ 0, Z \text{ and } N \text{ even} \end{cases}$$ ---------- the average pairing energy

$$-a_{el}Z^{2.39}\,.$$ ---------- the energy of bound electrons





Moller, Atomic Data and Nuclear Data Tables, 109-110

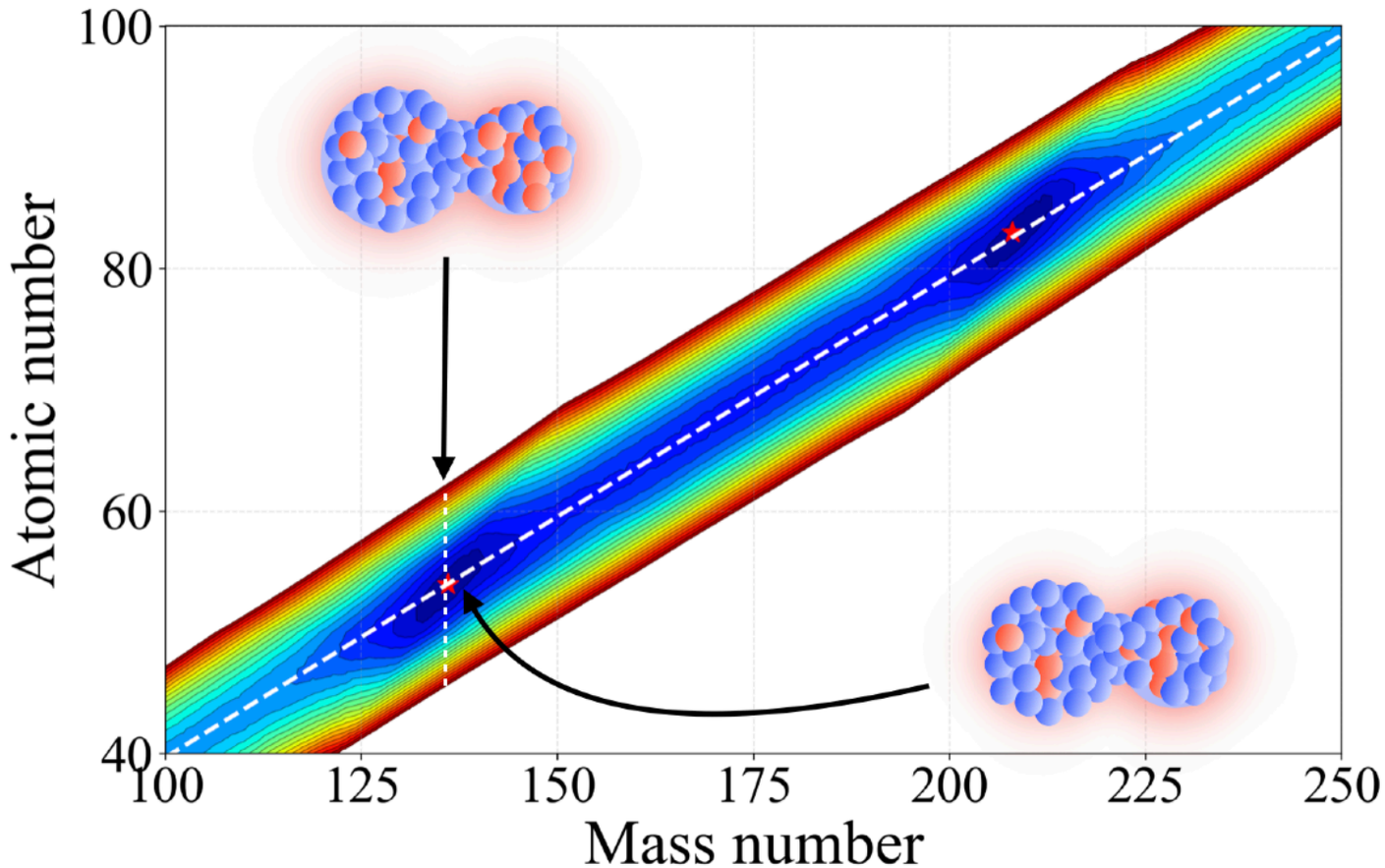
Potential energy surface - Adiabatic potential energy

In order to calculate the macroscopic mass for an arbitrary value of η_Z

$$M_{\mathrm{mac}}(A,Z;q) \simeq M_{\mathrm{mac}}(A,Z;\overrightarrow{q})\,|_{\eta_Z=\langle\eta_Z\rangle} + C_{\eta_Z}\frac{(\eta_Z-\langle\eta_Z\rangle)^2}{2},$$

where $\langle\eta_Z\rangle$ is the charge asymmetry providing the minimum of the potential energy and C_{η_Z} is the stiffness coefficient.

$$M_{\mathrm{mac}}(A,Z;q) \approx M_{\mathrm{mac}}(A,Z;\overrightarrow{q})\Big|_{n_Z=n_A} - C_{\eta_Z}\frac{(\eta_A-\langle\eta_Z\rangle)^2}{2} + C_{\eta_Z}\frac{(\eta_Z-\langle\eta_Z\rangle)^2}{2},$$



Potential energy surface - The Strutinsky method

The Hamiltonian of the TCSM in cylindrical coordinates can be written as

$$H = -\frac{\hbar^2 \nabla^2}{2m} + V(\rho, z) + V_{\mathrm{LS}}(r, l, s) + V_{\mathrm{L}^2}(r, l)$$

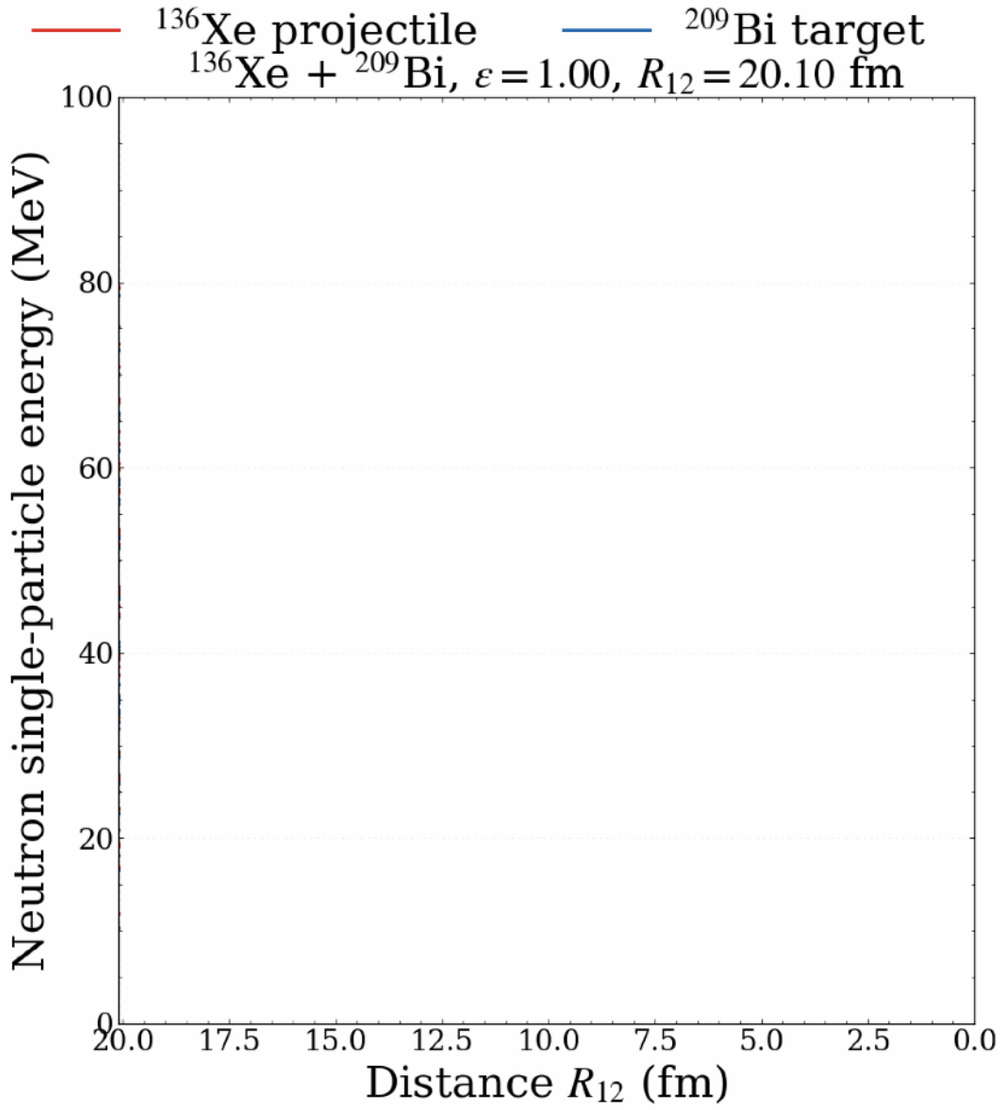
Two-center potential

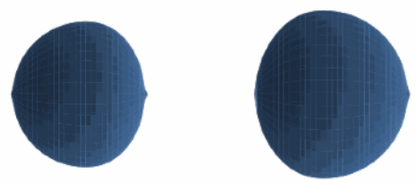
$$V_i(\rho, z) \simeq \frac{1}{2} m_0 \omega_{zi}^2 (z - z_i)^2 + \frac{1}{2} m_0 \omega_{\rho i}^2 \rho^2, \qquad i = 1,2.$$

For a given nuclear shape, the single-particle states are obtained from

$$\hat{H}\Phi_\alpha = \epsilon_\alpha \Phi_\alpha, \qquad \Phi(\rho, z, \phi) = f(z)\, g(\rho)\, h(\phi)\,.$$

so the basis states can be written as $|n_z n_\rho m\rangle$.

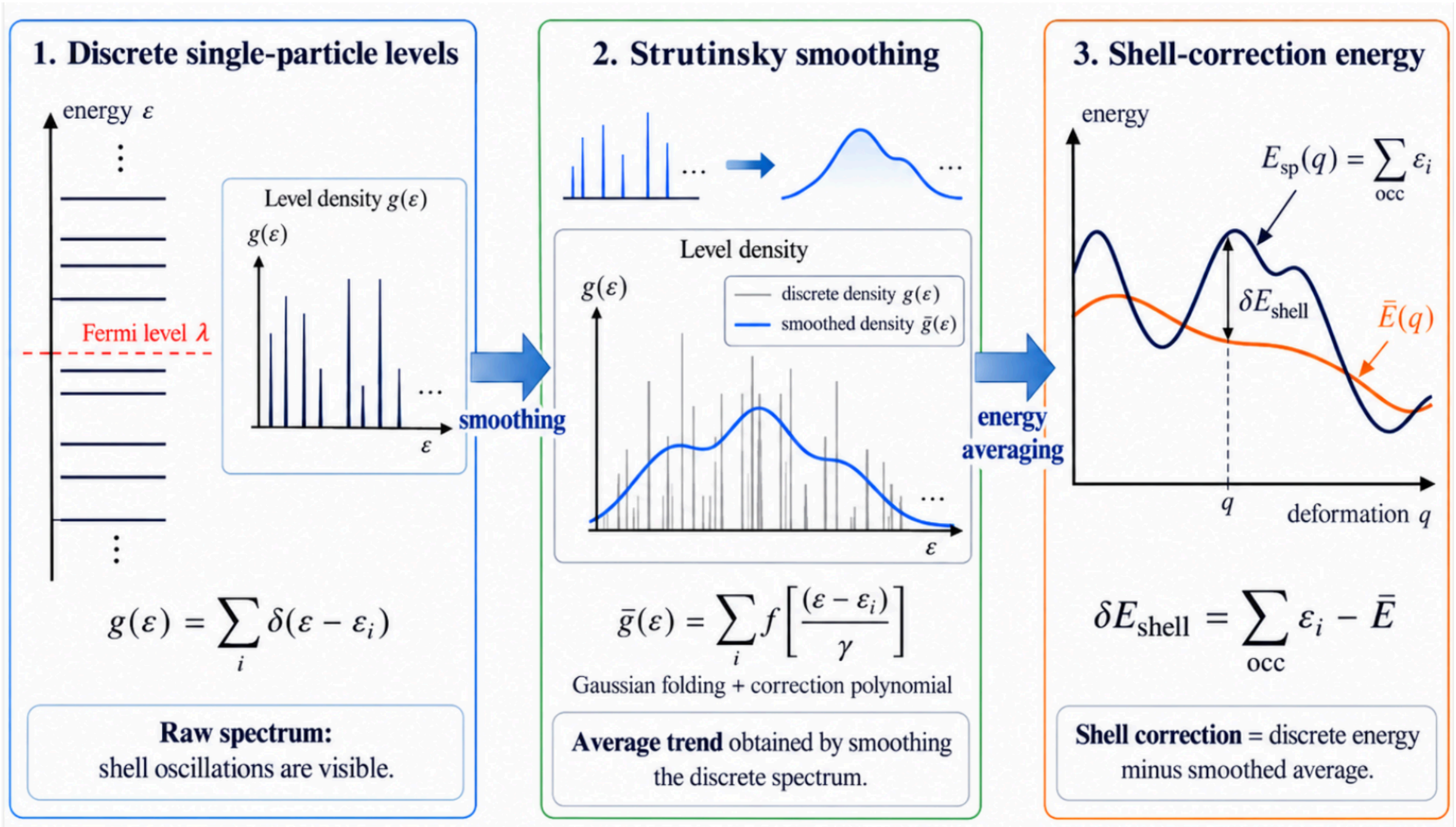




J. Maruhn and W. Greiner, Z. Phys. A 251, 431 (1972).

Potential energy surface - The Strutinsky method

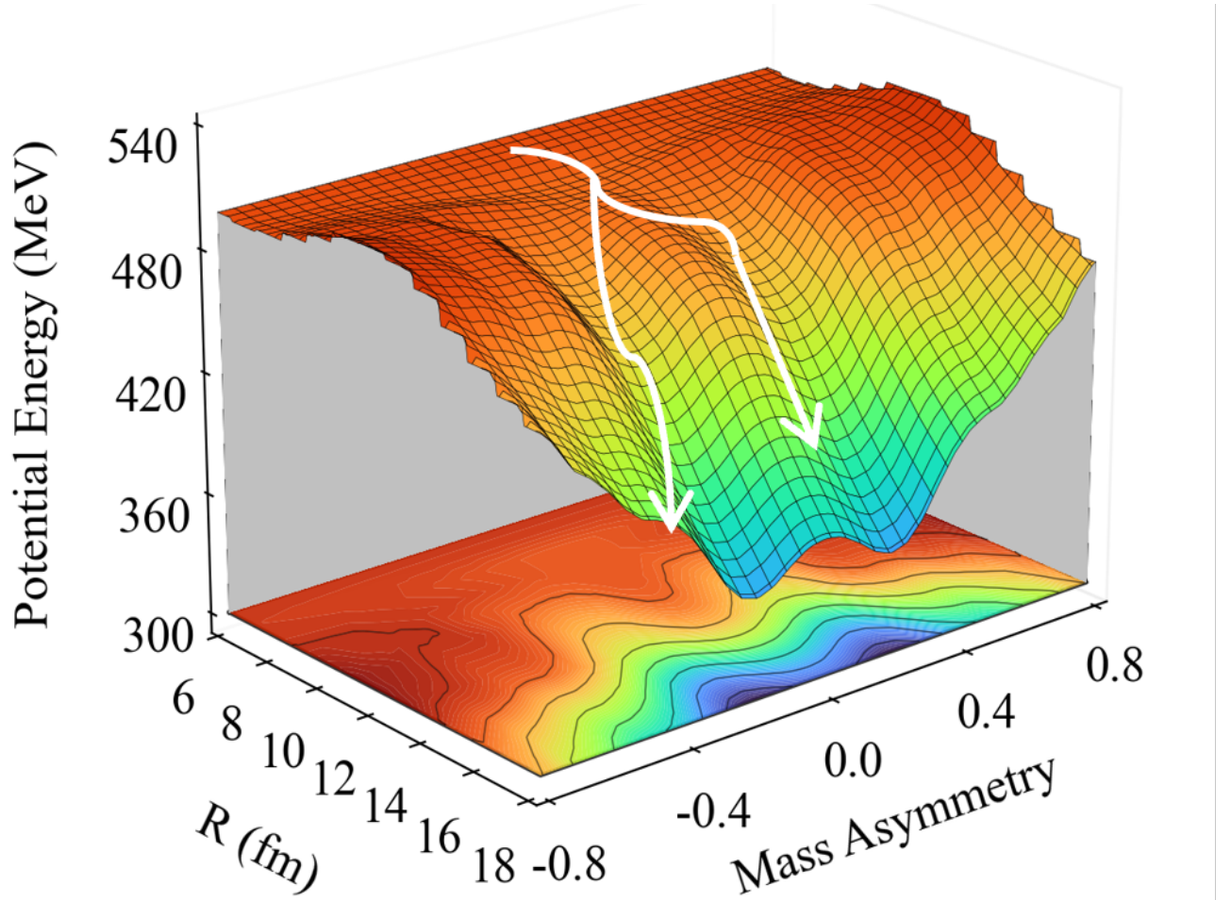
The shell correction reflects the deviation of single-particle energy levels from a uniform distribution.



Summarised by GPT

V. M. Strutinsky, Nucl. Phys. A 95, 420 (1967); 22, 1 (1968).

Equations of motion and nucleon transfer



Langevin equation

$$\dot{q}_i = \mu_{ij} p_j, \quad \dot{p}_i = F_i^{\text{driving}} + F_i^{\text{friction}} + F_i^{\text{random}}$$

$$q_i = \{r, \delta_1 = \delta_2, \theta, \varphi_1, \varphi_2\}$$

$$p_i = \{p_r, p_{\delta_1} = p_{\delta_2}, L, l_1, l_2\}$$

- **Describing single event shape evolution**

Master equation

$$\frac{\partial P(\mathbf{S}, t)}{\partial t} = \sum_{\mathbf{S}' \neq \mathbf{S}} W(\mathbf{S}', \mathbf{S}) P(\mathbf{S}', t) - W(\mathbf{S}, \mathbf{S}') P(\mathbf{S}, t)$$

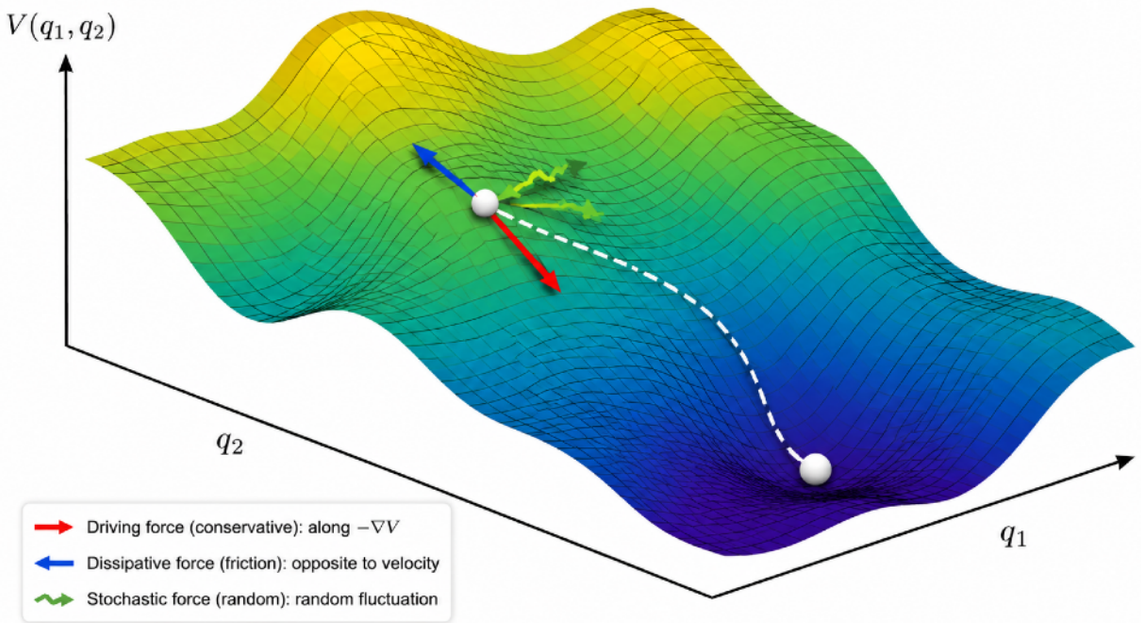
$$\mathbf{S} = \{N, Z\}$$

- **Describing nucleon transfer evolution**

Equations of motion and nucleon transfer

Modeling the time evolution of the nuclear system, we can use the Langevin equation

$$\dot{q}_i = \mu_{ij} p_j, \quad \dot{p}_i = F_i^{\text{driving}} + F_i^{\text{friction}} + F_i^{\text{random}}$$



$$\frac{dp_r}{dt} = -\frac{\partial V}{\partial r} - \sum_{j,k} \frac{p_j p_k}{2} \frac{\partial \mu_{jk}}{\partial r} + \frac{\hbar^2 l^2}{2\mathfrak{I}_{12}^2} \frac{\partial \mathfrak{I}_{12}}{\partial r} + \sum_{i=1,2} \frac{\hbar^2 l_i^2}{2\mathfrak{I}_i^2} \frac{\partial \mathfrak{I}_i}{\partial r} - \sum_{j,k} \gamma_{rj} \mu_{jk} p_k + \sum_j g_{rj} \Gamma_j$$

$$\frac{dp_{\delta_1}}{dt} = -\frac{\partial V}{\partial \delta_1} - \sum_{j,k} \frac{p_j p_k}{2} \frac{\partial \mu_{jk}}{\partial \delta_1} + \frac{\hbar^2 l^2}{2\mathfrak{I}_{12}^2} \frac{\partial \mathfrak{I}_{12}}{\partial \delta_1} + \frac{\hbar^2 l_1^2}{2\mathfrak{I}_1^2} \frac{\partial \mathfrak{I}_1}{\partial \delta_1} - \sum_{j,k} \gamma_{\delta_1 j} \mu_{jk} p_k + \sum_j g_{\delta_1 j} \Gamma_j$$

$$\frac{dp_{\delta_2}}{dt} = -\frac{\partial V}{\partial \delta_2} - \sum_{j,k} \frac{p_j p_k}{2} \frac{\partial \mu_{jk}}{\partial \delta_2} + \frac{\hbar^2 l^2}{2\mathfrak{I}_{12}^2} \frac{\partial \mathfrak{I}_{12}}{\partial \delta_2} + \frac{\hbar^2 l_2^2}{2\mathfrak{I}_2^2} \frac{\partial \mathfrak{I}_2}{\partial \delta_2} - \sum_{j,k} \gamma_{\delta_2 j} \mu_{jk} p_k + \sum_j g_{\delta_2 j} \Gamma_j$$

$$\frac{dl}{dt}\hbar = -\frac{\partial V}{\partial \theta} - \gamma_{\text{tang}} \left(\frac{\hbar l}{\mathfrak{I}_{12}} R_{\text{c.m.}} - \frac{\hbar l_1}{\mathfrak{I}_1} R_{\text{c.m.}}^{(1)} - \frac{\hbar l_2}{\mathfrak{I}_2} R_{\text{c.m.}}^{(2)} \right) R_{\text{c.m.}} + R_{\text{c.m.}} \sqrt{\gamma_{\text{tang}} T} \Gamma_{\text{tang}}$$

$$\frac{dl_1}{dt}\hbar = -\frac{\partial V}{\partial \varphi_1} + \gamma_{\text{tang}} \left(\frac{\hbar l}{\mathfrak{I}_{12}} R_{\text{c.m.}} - \frac{\hbar l_1}{\mathfrak{I}_1} R_{\text{c.m.}}^{(1)} - \frac{\hbar l_2}{\mathfrak{I}_2} R_{\text{c.m.}}^{(2)} \right) R_{\text{c.m.}}^{(1)} - R_{\text{c.m.}}^{(1)} \sqrt{\gamma_{\text{tang}} T} \Gamma_{\text{tang}}$$

$$\frac{dl_2}{dt}\hbar = -\frac{\partial V}{\partial \varphi_2} + \gamma_{\text{tang}} \left(\frac{\hbar l}{\mathfrak{I}_{12}} R_{\text{c.m.}} - \frac{\hbar l_1}{\mathfrak{I}_1} R_{\text{c.m.}}^{(1)} - \frac{\hbar l_2}{\mathfrak{I}_2} R_{\text{c.m.}}^{(2)} \right) R_{\text{c.m.}}^{(2)} - R_{\text{c.m.}}^{(2)} \sqrt{\gamma_{\text{tang}} T} \Gamma_{\text{tang}}$$

friction from the wall-and-window one-body dissipation model
collective inertia from a hydrodynamical approach using the Werner-Wheeler flow field
The random force is linked to friction by a fluctuation-dissipation relation

Equations of motion and nucleon transfer

A master equation is a phenomenological set of first-order differential equations describing the time evolution of (usually) the probability of a system to occupy each one of a discrete set of states with regard to a continuous time variable t.

$$\frac{\partial P(\mathbf{S},t)}{\partial t} = \sum_{\mathbf{S}'\neq\mathbf{S}} W(\mathbf{S}',\mathbf{S})P(\mathbf{S}',t) - W(\mathbf{S},\mathbf{S}')P(\mathbf{S},t)$$

The transition probability from the state to another state

Macroscopic transition probabilities

Microscopic transition probabilities

$$W^{\pm} = \lambda_0 \cdot D(q,S) \cdot P_{tr}(q, S \rightarrow S \pm 1)$$

Diffusion constant

Probability of a nucleon transfer

V. Zagrebaev, A. Karpov, Y. Aritomo, M. Naumenko, and W. Greiner, Phys. Part. Nucl. 38, 469 (2007).

The transition probability from the state to another state

Macroscopic transition probabilities

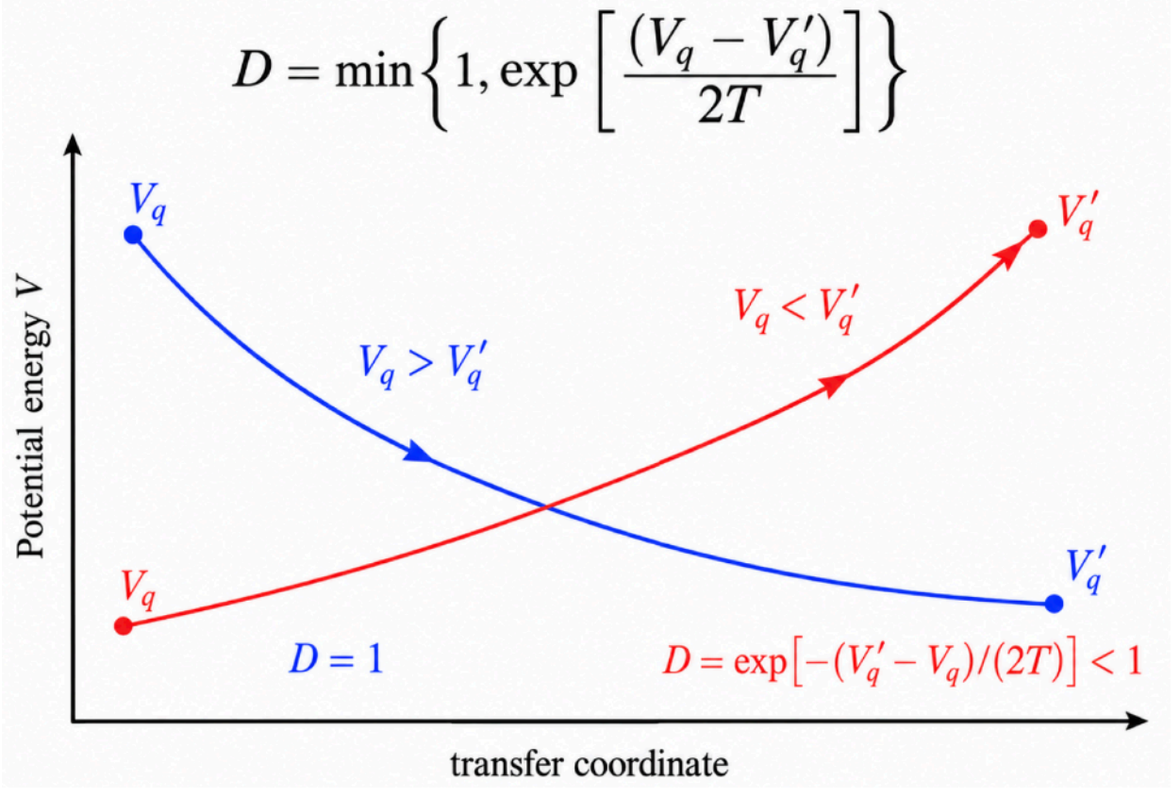
Microscopic transition probabilities

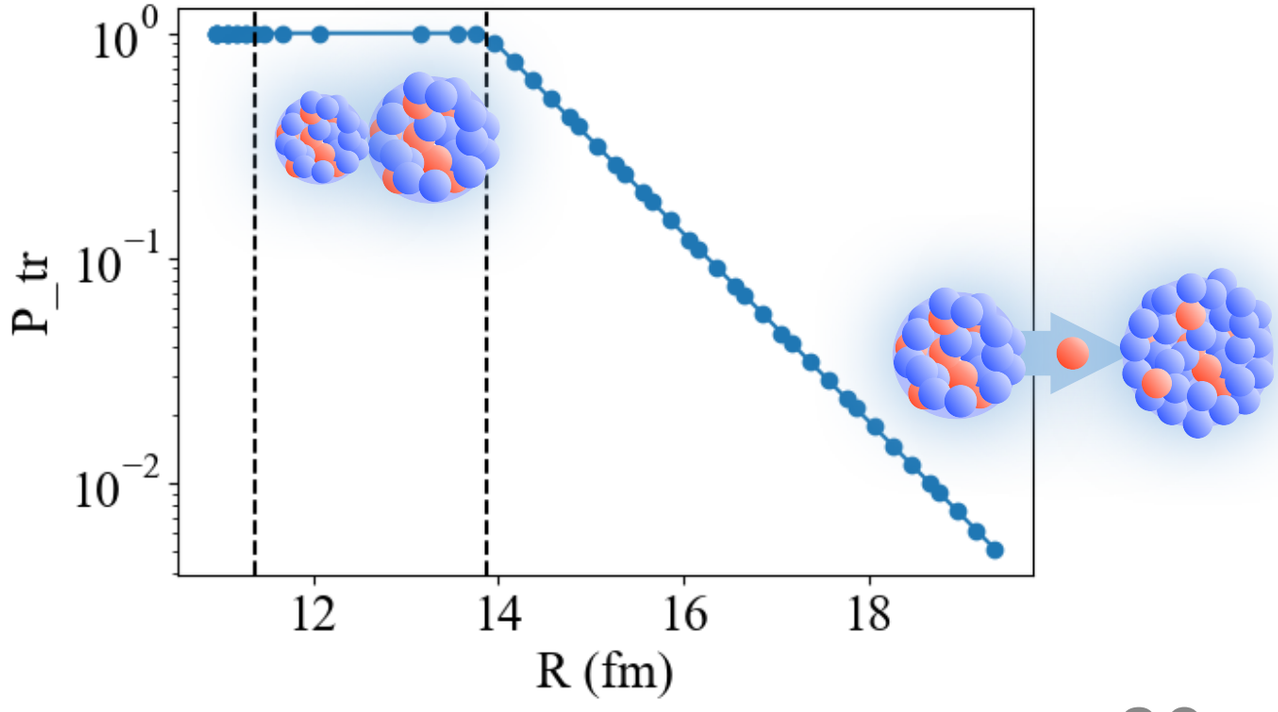
$$W^{\pm} = \lambda_0 \cdot D(q,S) \cdot P_{tr}(q, S \to S \pm 1)$$

Diffusion constant

Probability of a nucleon transfer

$$D = \min\left\{1, \exp\left[\frac{(V_q - V_q')}{2T}\right]\right\}$$



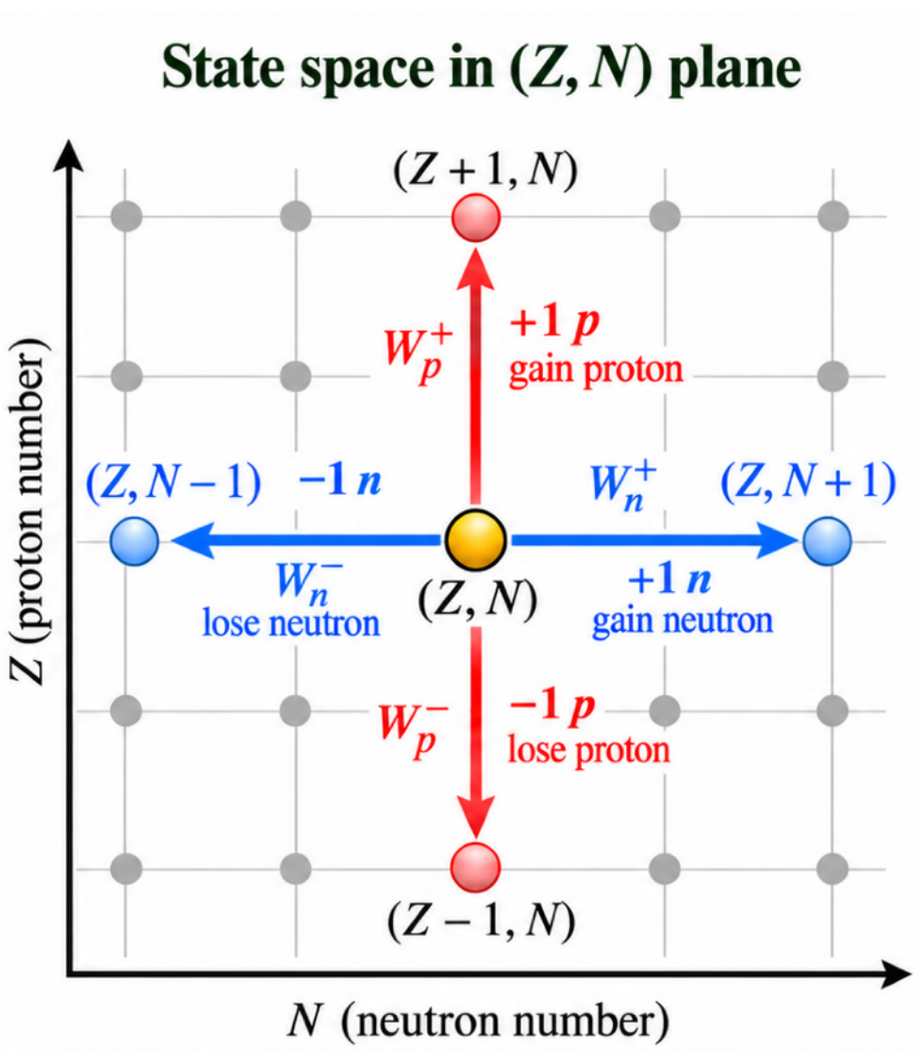


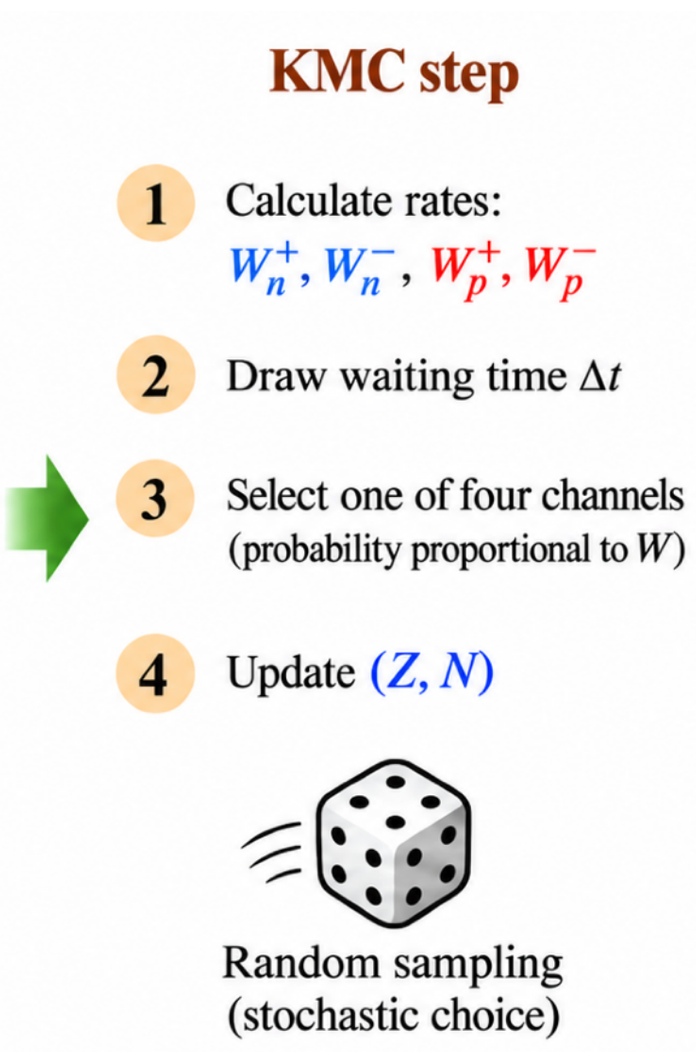
$$P_{tr}(R, \beta, A \to A \pm 1) = \exp(-2k[r_c - r_0])$$

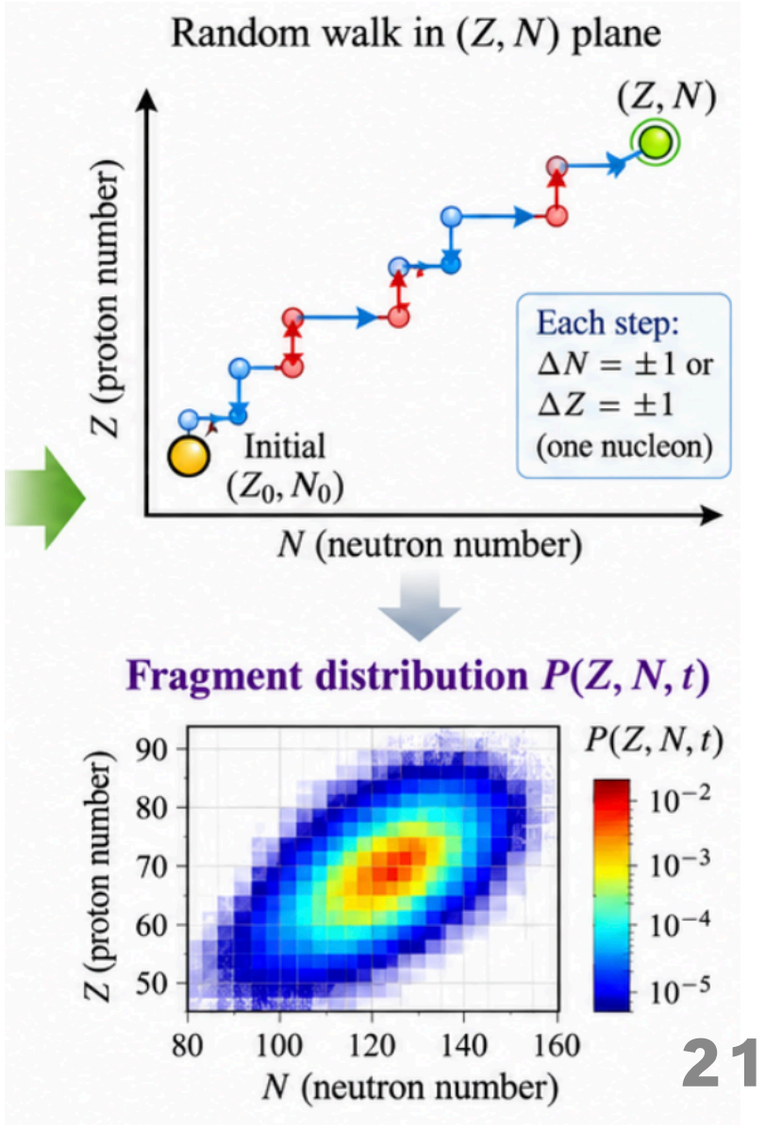
Equations of motion and nucleon transfer

A master equation is a phenomenological set of first-order differential equations describing the time evolution of (usually) the probability of a system to occupy each one of a discrete set of states with regard to a continuous time variable t.

$$\frac{\partial P(\mathbf{S},t)}{\partial t} = \sum_{\mathbf{S}'\neq\mathbf{S}} W(\mathbf{S}',\mathbf{S})P(\mathbf{S}',t) - W(\mathbf{S},\mathbf{S}')P(\mathbf{S},t)$$

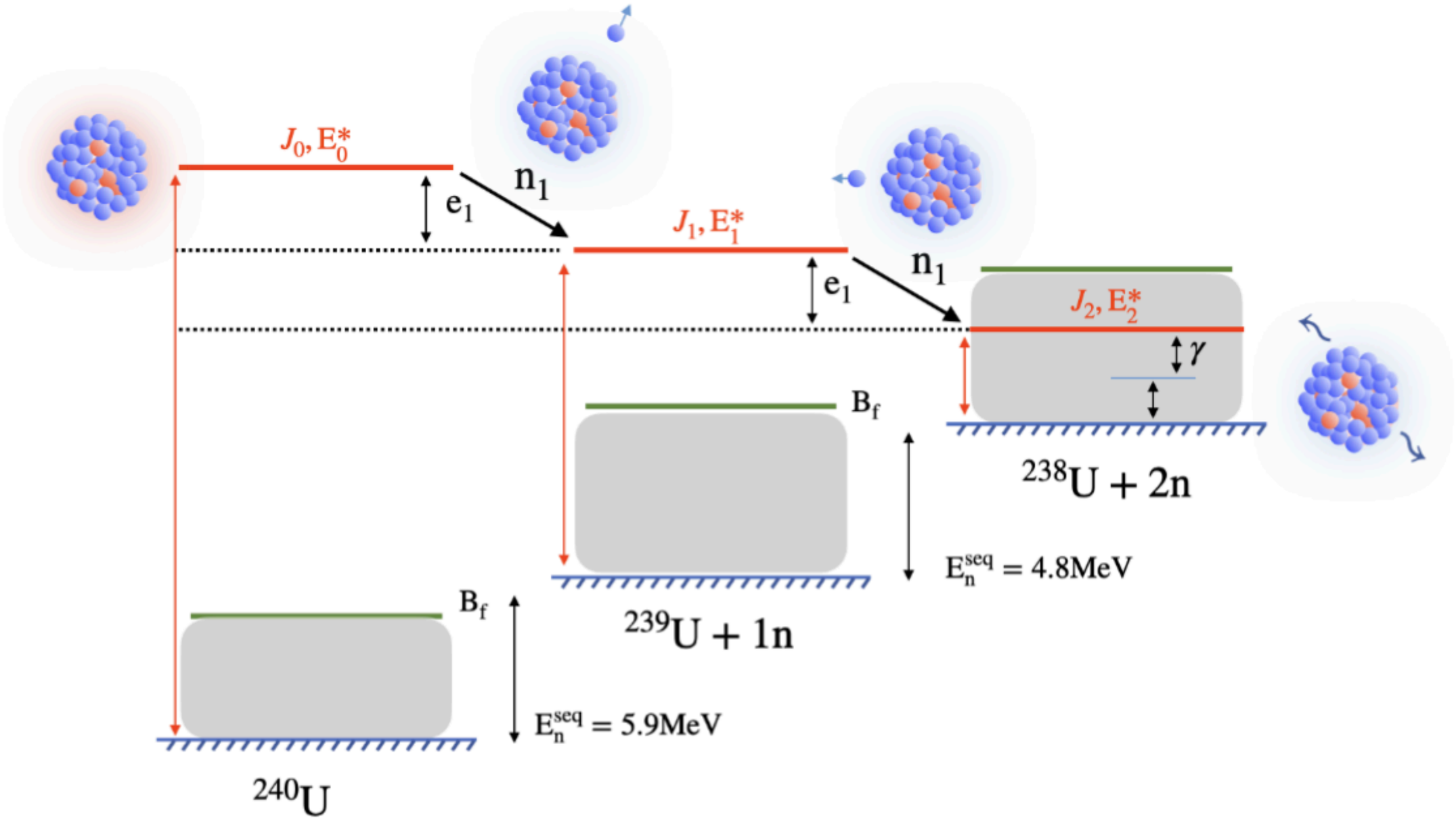






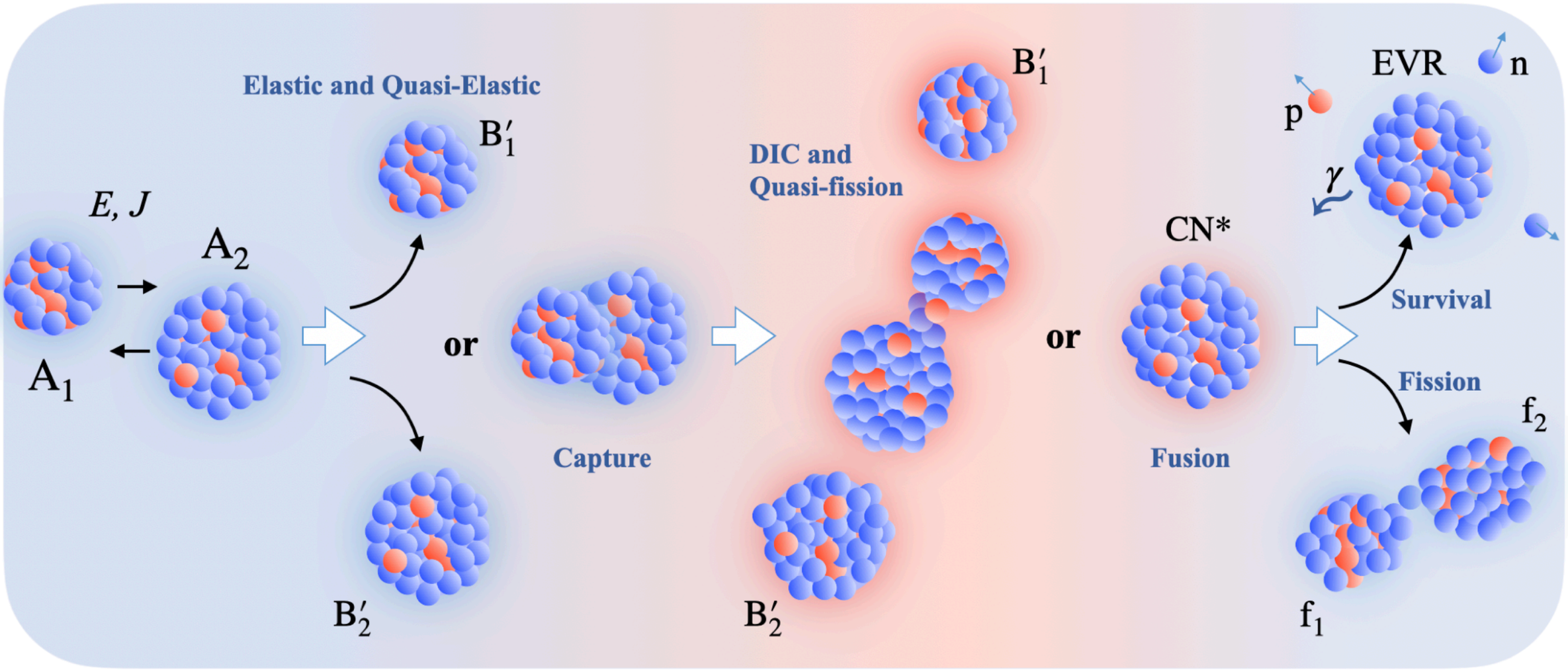
Modeling nucleus-nucleus collisions

Statistical Model GEMINI++ has been used



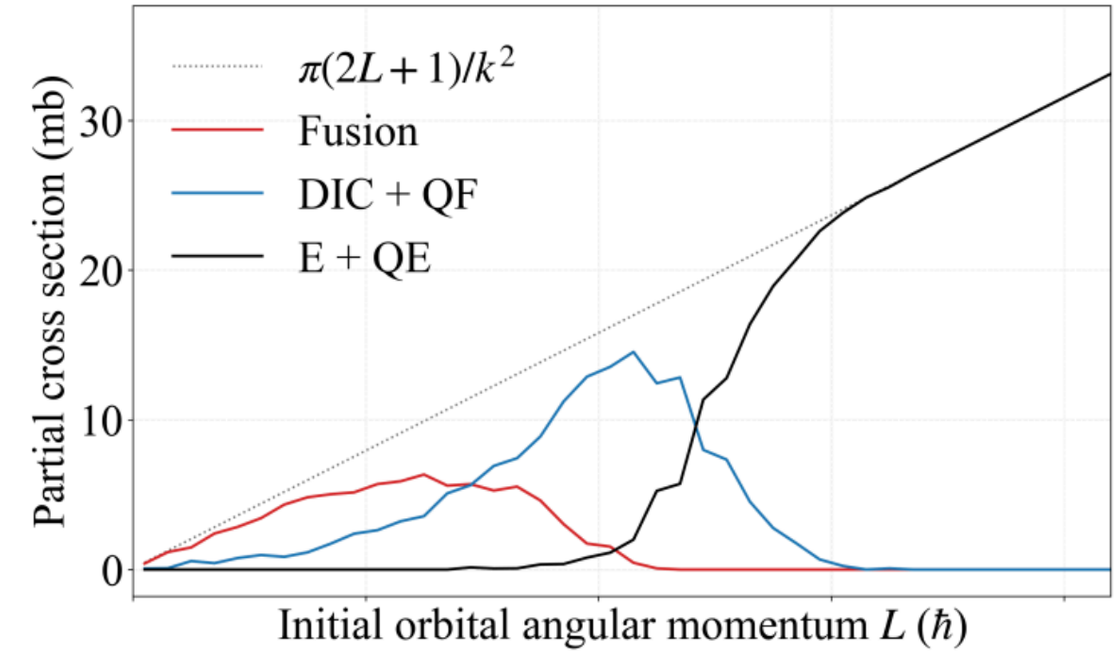
R. J. Charity, PHYSICAL REVIEW C 82, 014610 (2010)

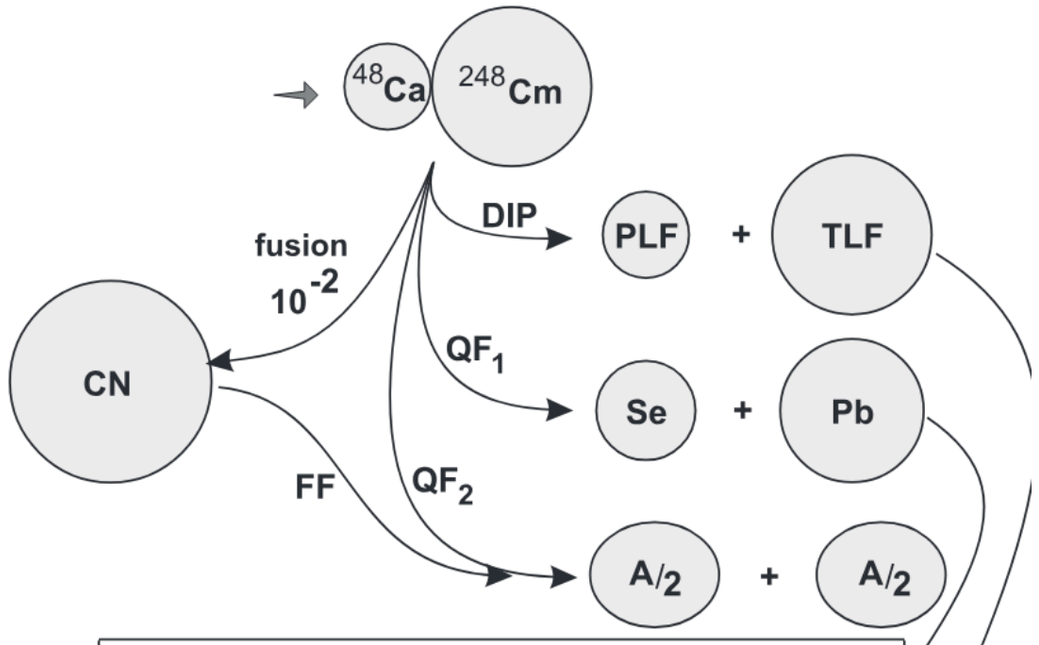
Modeling nucleus-nucleus collisions

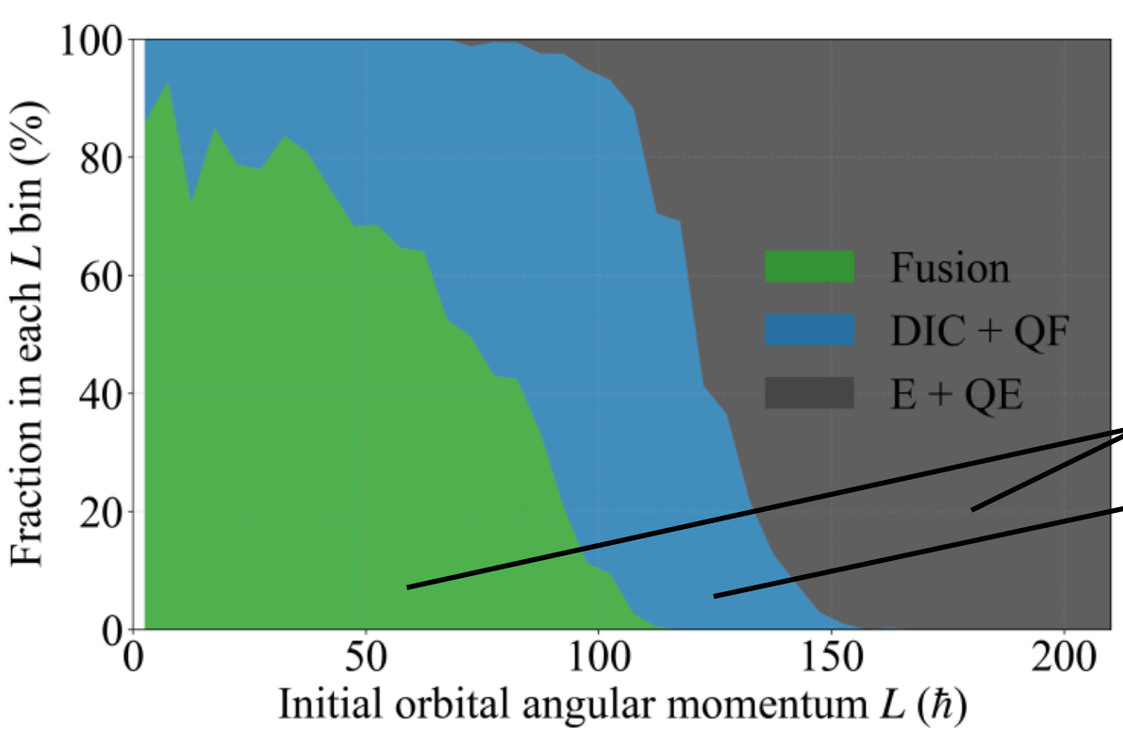


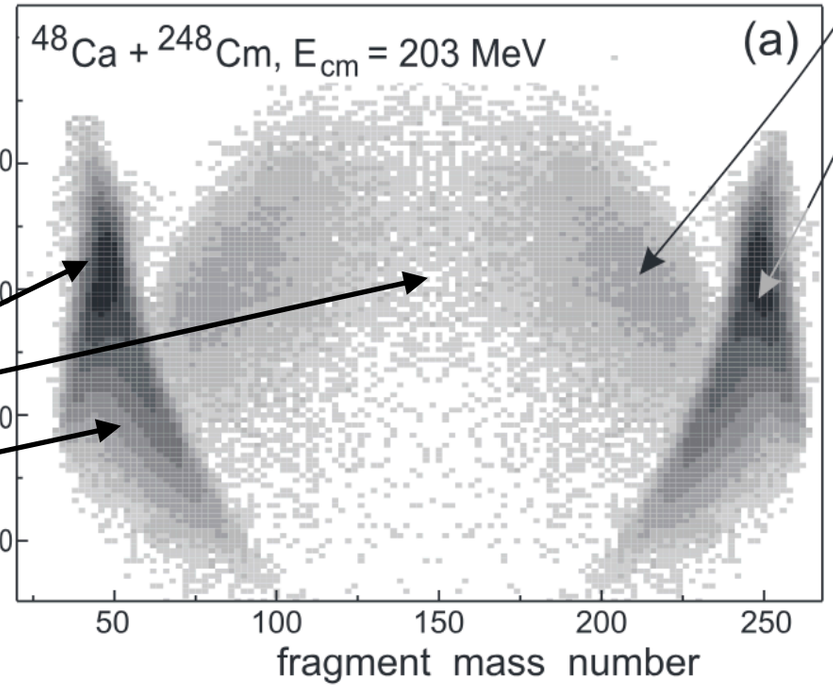
$$\sigma\left(Z_f, N_f\right) = \frac{\pi\hbar^2}{2\mu E_{\text{c.m.}}} \sum_{J=0}^{J_{\max}} (2J+1) \sum_{Z_1,N_1} P(Z_1, N_1; J^*, E^*) W(Z_1, N_1 \to Z_f, N_f; J^*, E^*),$$

Modeling nucleus-nucleus collisions









Modeling nucleus-nucleus collisions

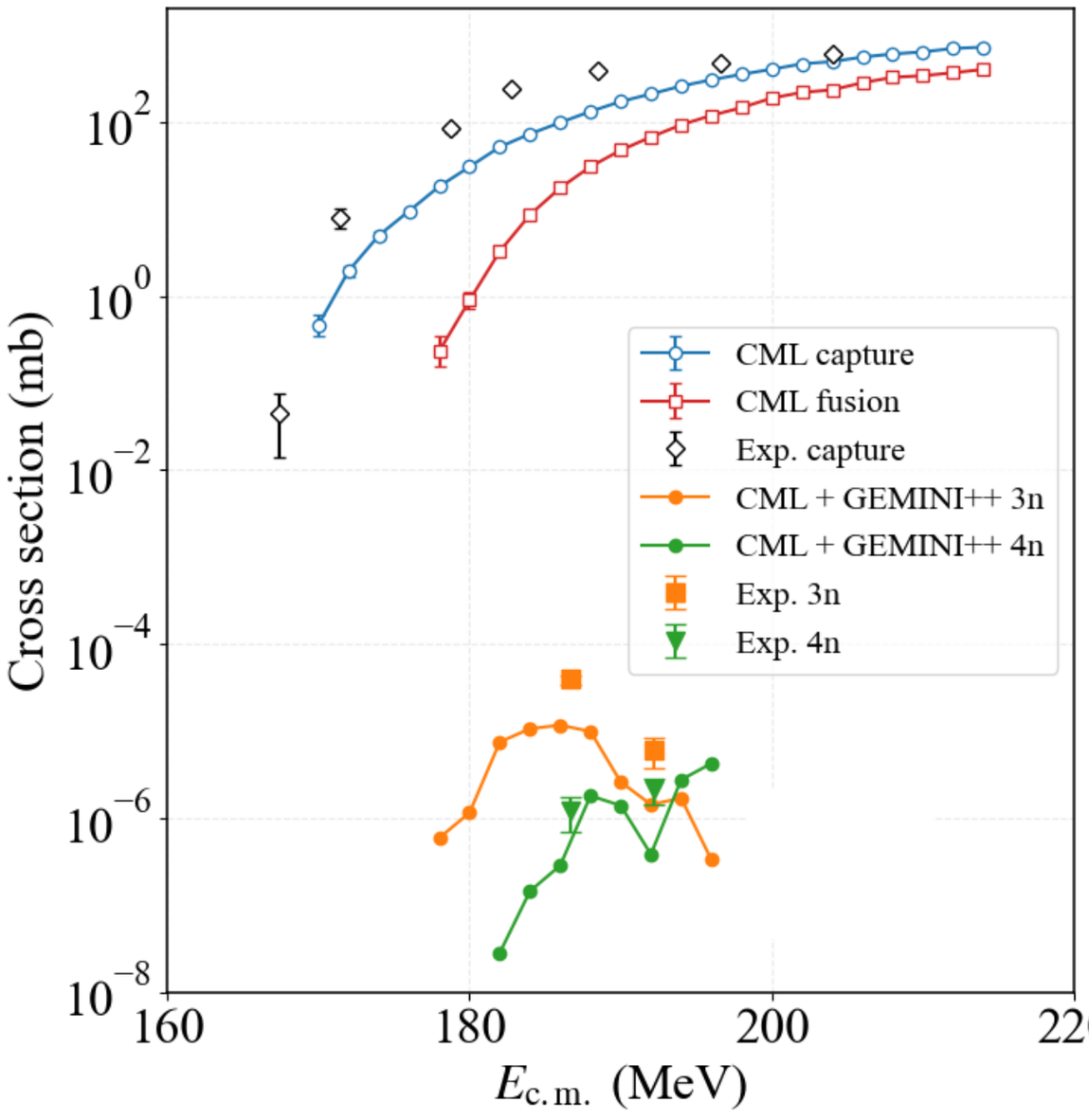
^{48}Ca + ^{208}Pb

Capture and non-Capture:

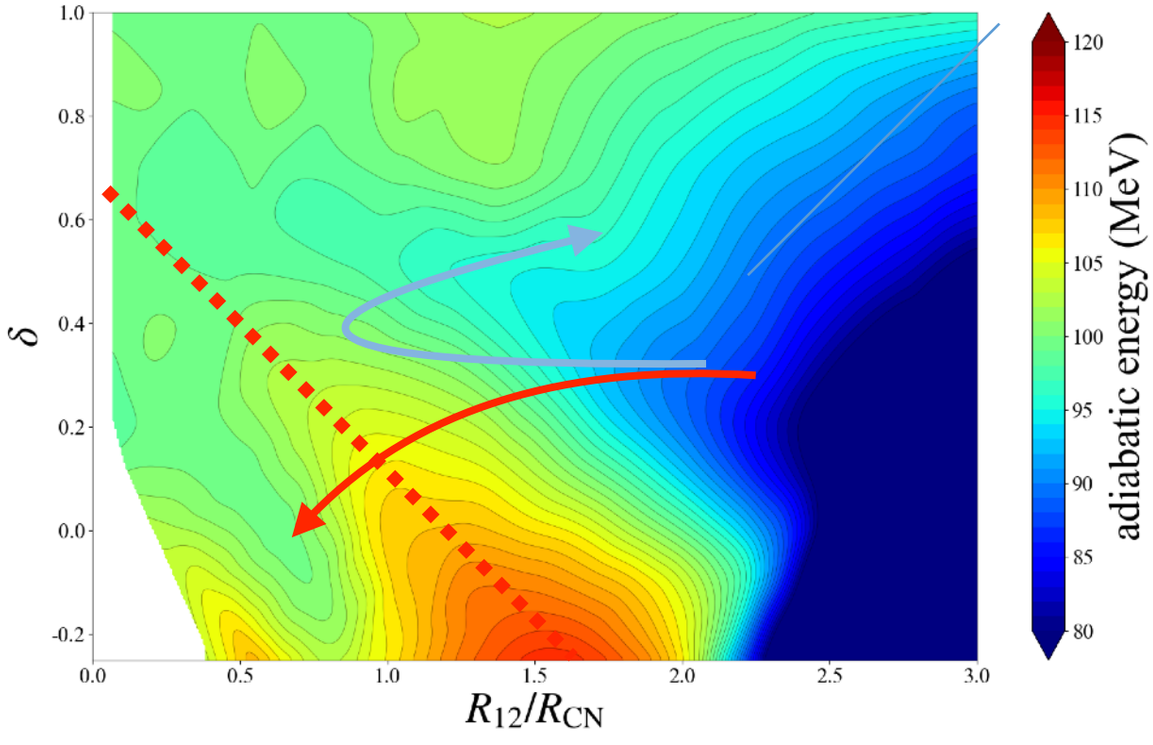
$$\sigma_x(E) = 2\pi \int_0^{b_{\max}} b\, P_x(E,b)\, db$$

$$P_x(E,b) = \frac{N_x(E,b)}{N_{\rm tot}(E,b)}, \qquad x = \text{cap, fus}$$





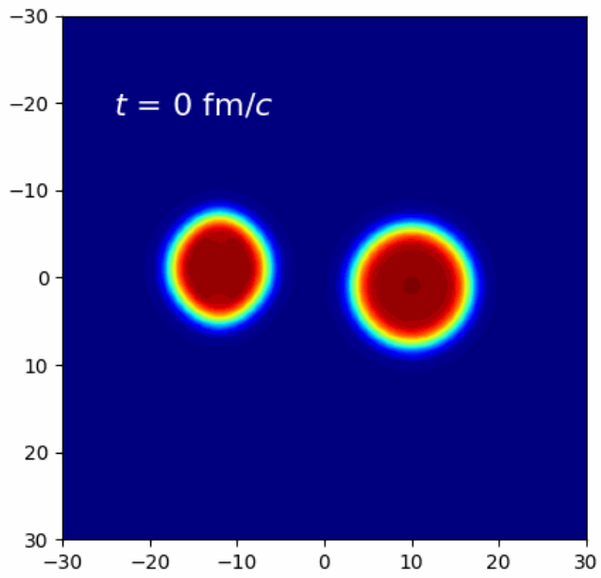
Fusion and non-Fusion:



Y. Aritomo, K. Hagino, K. Nishio, and S. Chiba, PHYSICAL REVIEW C 85, 044614 (2012)

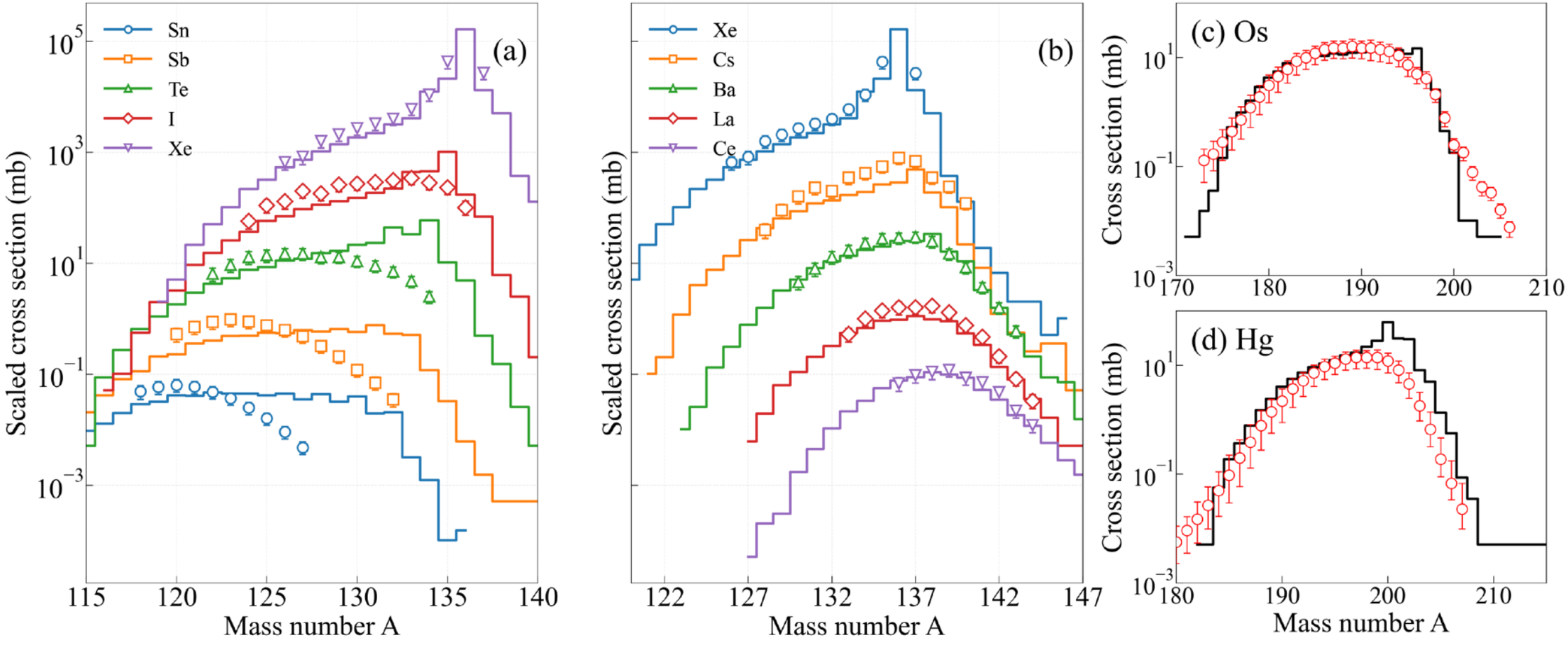
Modeling nucleus-nucleus collisions $^{136}\mathrm{Xe} + {}^{198}\mathrm{Pt}$

In conclusion, the promised potential of the production of new isotopes around and beyond the neutron shell $N = 126$ by multinucleon transfer reactions was established for the first time in the $^{136}\mathrm{Xe} + {}^{198}\mathrm{Pt}$ system at an energy above the Coulomb barrier. The absolute cross



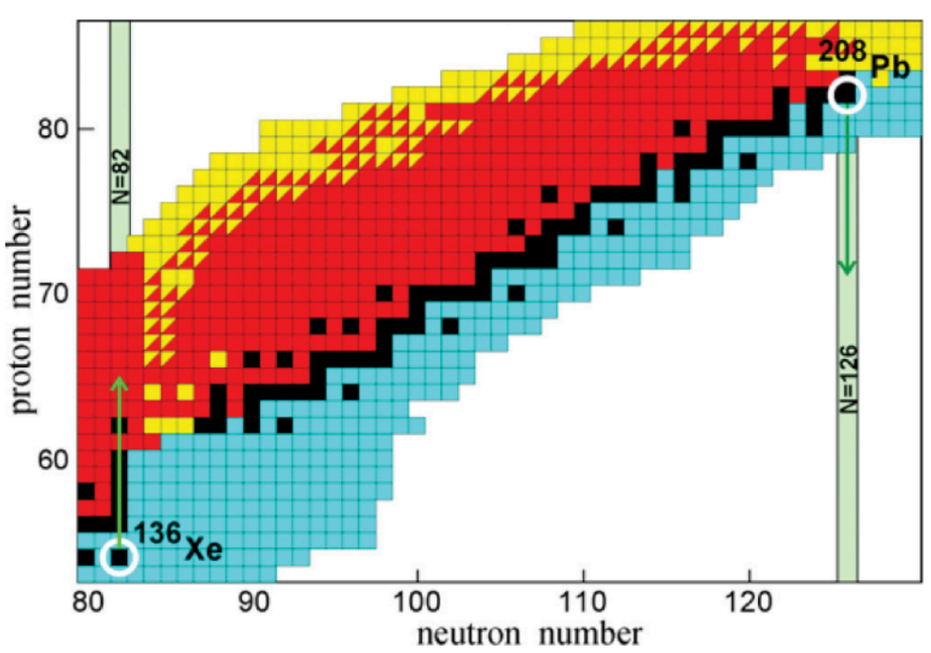
Y. X. Watanabe et al., PRL 115, 172503 (2015)

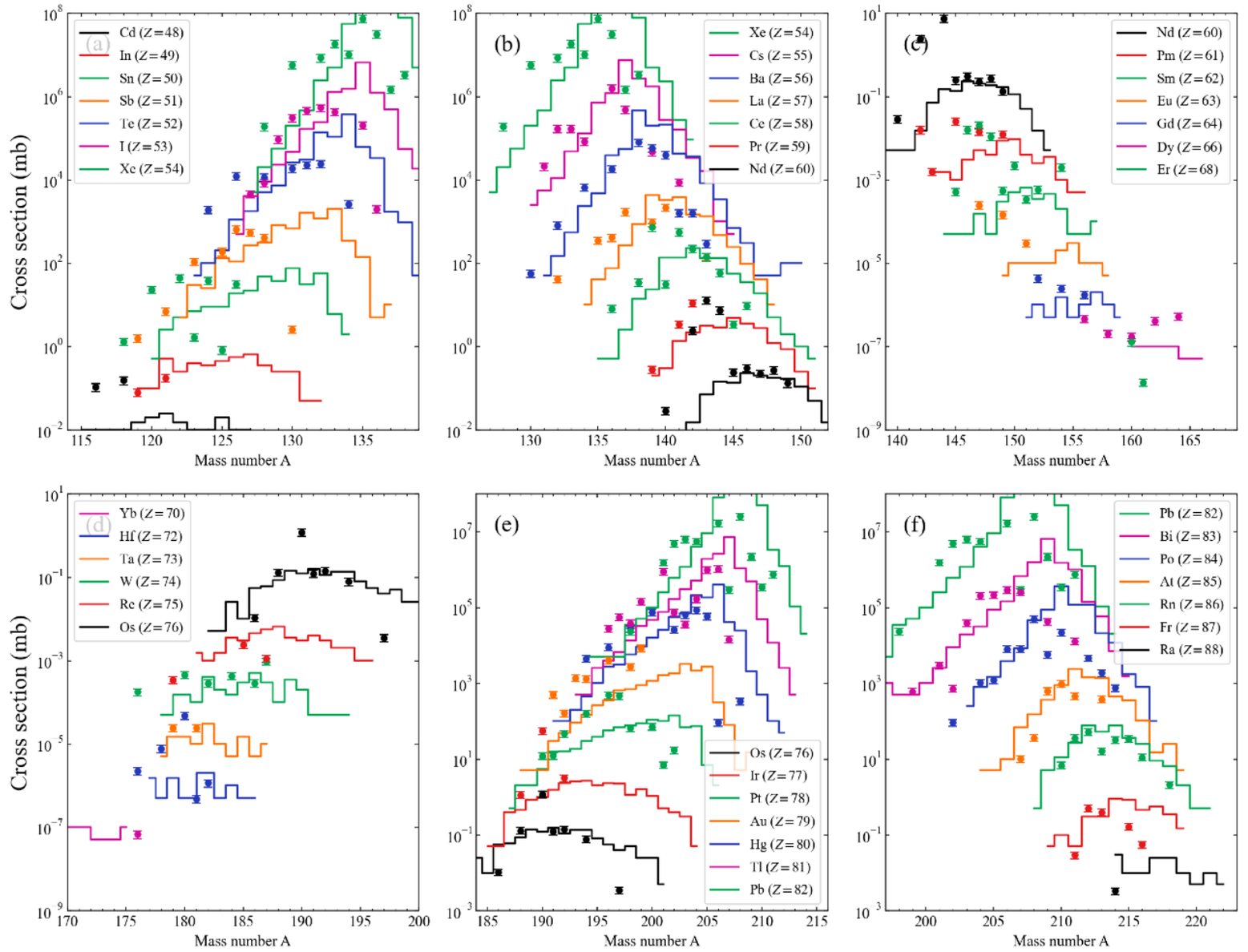
Y. X. Watanabe et al.,
Nucl. Instrum. Methods Phys. Res., Sect. B 317, 752 (2013).



Modeling nucleus-nucleus collisions $^{136}\mathrm{Xe} + {}^{208}\mathrm{Pb}$

According to this model, the reaction $^{136}\mathrm{Xe} + {}^{208}\mathrm{Pb}$ at energies around the Coulomb barrier, is a good candidate for the production of new neutron-rich isotopes in the region of the neutron shell closure $N = 126$. Calculations demonstrate that the production of a few tens of unknown neutron-rich target-like fragments (TLF) in the mass region $A \approx 200$ is possible with cross sections of not less than 1 μb if the multinucleon transfer occurs. It is worth to remark that the multinucleon transfer process conjectured in [3] is more favorable with respect to the case of the cold fragmentation reaction $^{208}\mathrm{Pb}(1A\ \mathrm{GeV}) + \mathrm{Be}$ [7], where nuclei in the same

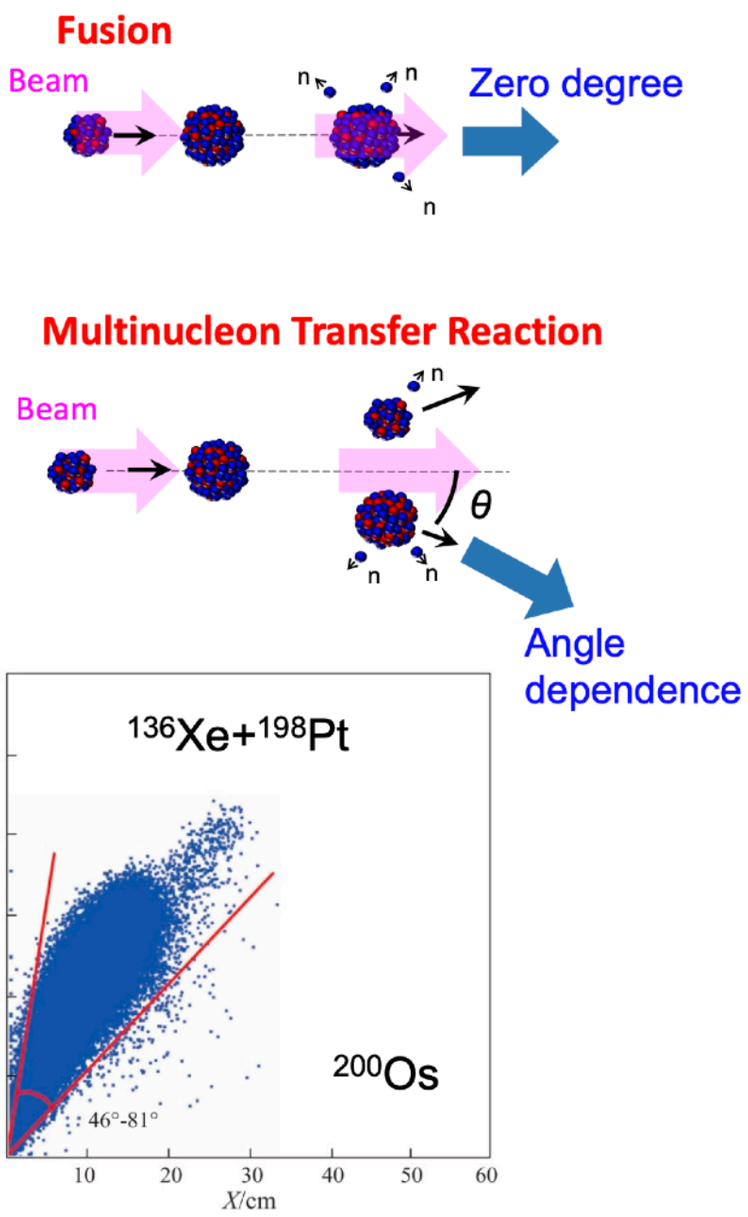


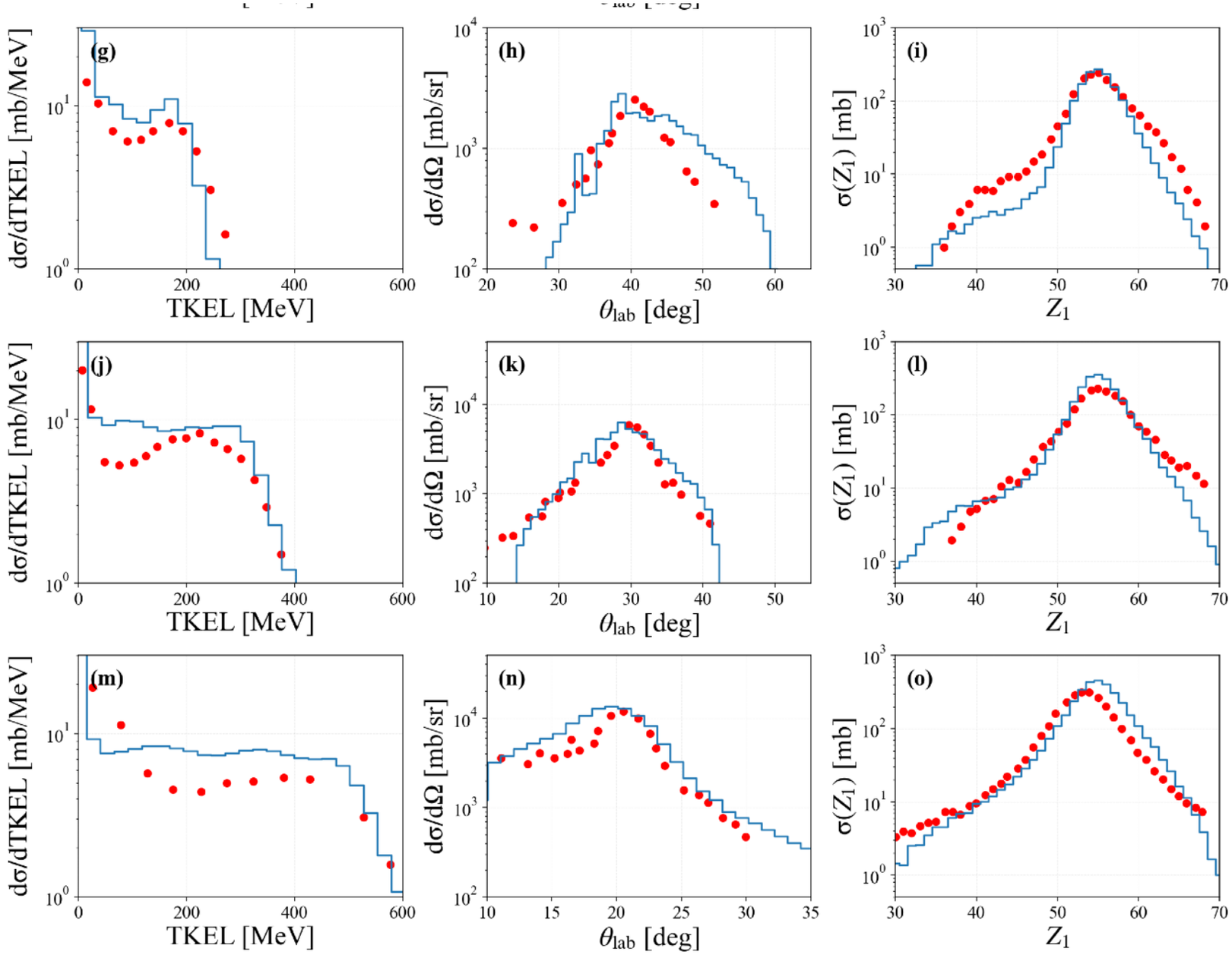


PHYSICAL REVIEW C 86, 044611 (2012)

Modeling nucleus-nucleus collisions

$^{136}Xe + {}^{209}Bi$





Wenxue Huang et al. Nucl. Phys. Rev. (2017)

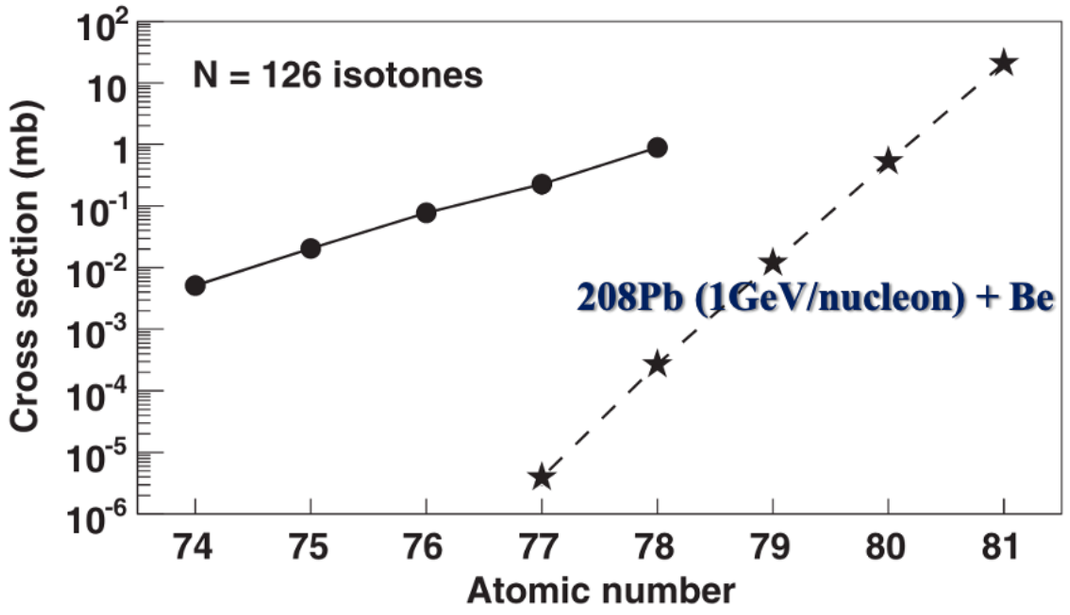
MNT · How to expand the nuclide chart

PRL 115, 172503 (2015) PHYSICAL REVIEW LETTERS week ending 23 OCTOBER 2015

Pathway for the Production of Neutron-Rich Isotopes around the $N = 126$ Shell Closure

Y. X. Watanabe,[1,*] Y. H. Kim,[2,3,†] S. C. Jeong,[1,‡] Y. Hirayama,[1] N. Imai,[1,§] H. Ishiyama,[1,‡] H. S. Jung,[1] H. Miyatake,[1] S. Choi,[2,3] J. S. Song,[2,3,4] E. Clement,[5] G. de France,[5] A. Navin,[5,‖] M. Rejmund,[5] C. Schmitt,[5] G. Pollarolo,[6] L. Corradi,[7] E. Fioretto,[7] D. Montanari,[8] M. Niikura,[9,¶] D. Suzuki,[9,**] H. Nishibata,[10] and J. Takatsu[10]

[1]*Institute of Particle and Nuclear Studies, High Energy Accelerator Research Organization (KEK), Tsukuba, Ibaraki 305-0801, Japan*

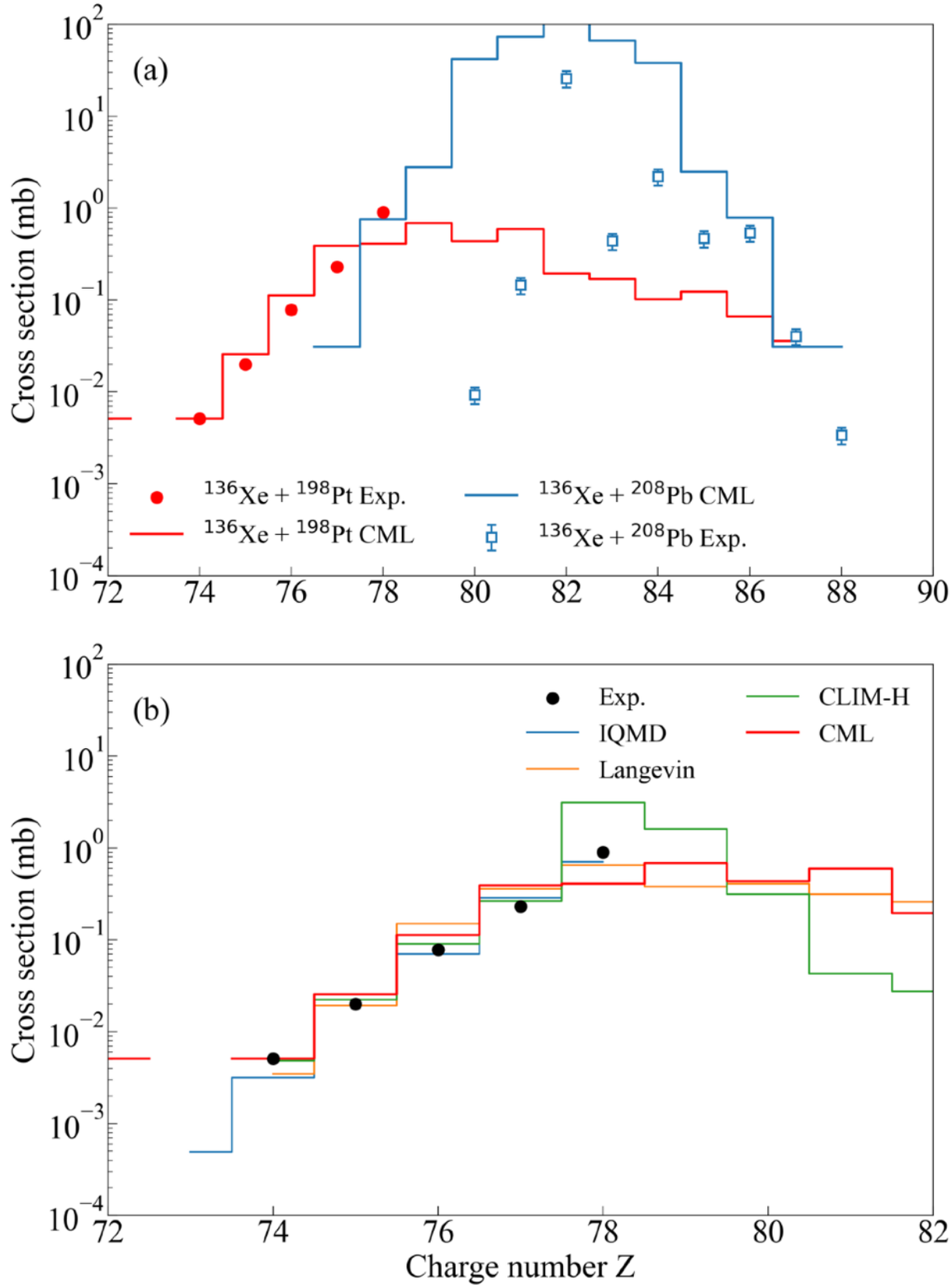


In conclusion, the promised potential of the production of new isotopes around and beyond the neutron shell $N = 126$ by multinucleon transfer reactions was established for the first time in the $^{136}Xe + ^{198}Pt$ system at an energy above the Coulomb barrier. The absolute cross

Y. X. Watanabe et al., PRL 115, 172503 (2015)

Y. X. Watanabe et al.,
Nucl. Instrum. Methods Phys. Res., Sect. B 317, 752 (2013).

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TDHF: I am here

Summary

- In this work, the coupled master–Langevin (CML) model has been developed to describe fusion and multinucleon transfer within a common dynamical framework.
- The calculations further support 136Xe+198Pt as a favorable reaction for producing neutron-rich N = 126 isotones.

Thank you for listening.

ご清聴ありがとうございました。

Potential energy surface - Adiabatic potential energy

The standard macro-microscopic model is often used for calculation of the total mass:

$$M(A,Z;q) = M_{\mathrm{mac}}(A,Z;q) + \delta E(A,Z;q)\,.$$

- $M_{\mathrm{mac}}(A,Z;q)$ **is macroscopic liquid-drop mass**
- $\delta E(A,Z;q)$ **is the microscopic shell correction by Strutinsky shell-correction method**

Nucleus	E_{mac}	E_{mic}	Mass	$E_{\mathrm{mac}}^{\mathrm{Moller}}$	$E_{\mathrm{mic}}^{\mathrm{Moller}}$	$\mathrm{Mass}^{\mathrm{Moller}}$	$\mathrm{Mass}^{\mathrm{exp}}$
^{40}Ca	-37.59	-5.12	42.71	-37.59	2.25	-35.34	-34.85
^{48}Ca	-43.54	-1.59	45.13	-43.54	0.11	-43.43	-44.21
^{132}Sn	-65.14	-12.32	77.46	-65.15	-11.55	-76.70	-76.61
^{136}Xe	-80.43	-6.86	87.29	-80.44	-7.20	-87.64	-86.43
^{208}Pb	-8.72	-11.82	20.54	-8.74	-12.84	-21.58	-21.77

the standard macro-microscopic approach and, in particular, to the case of highly deformed mononucleus and/or two separated nuclei leads to an incorrect result

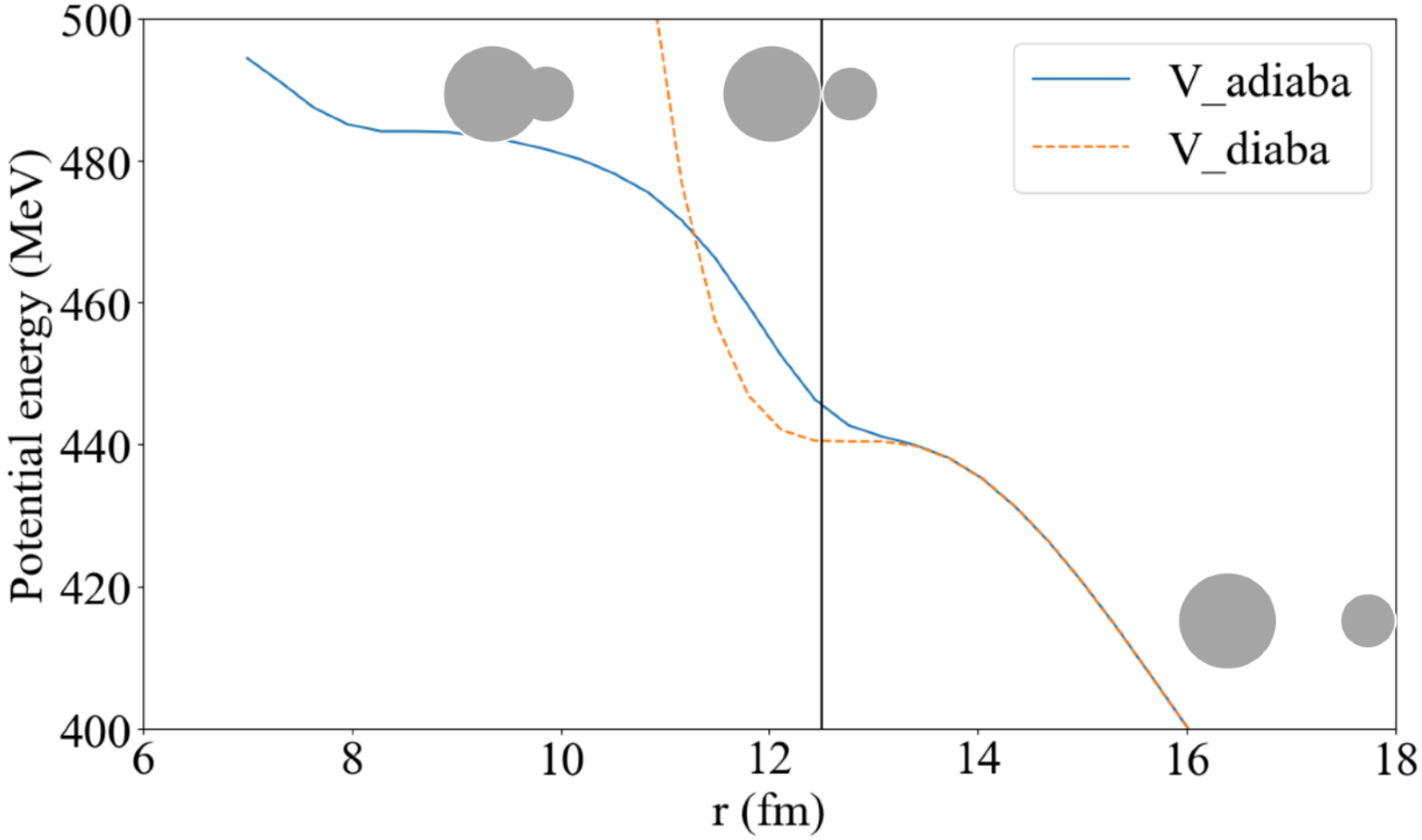
Potential energy surface - Adiabatic potential energy

In particular, the adiabatic potential energy can be calculated as follows:

$$V_{\text{adiab}}(A,Z;q) = V^{\text{standard}}_{\text{standard}}(A,Z;q)B(q) + V_{\text{diab}}(A,Z;q)[1-B(q)]$$

$$B(q) = [1+\exp(\frac{r-R_{\text{contact}}}{a_{\text{diff}}})]^{-2}, \quad a_{\text{diff}} = 0.5\text{fm}$$

where $V^{\text{standard}}_{\text{standard}}(A,Z;q)$ is the adiabatic potential energy calculated within the standard macro-microscopical approach.



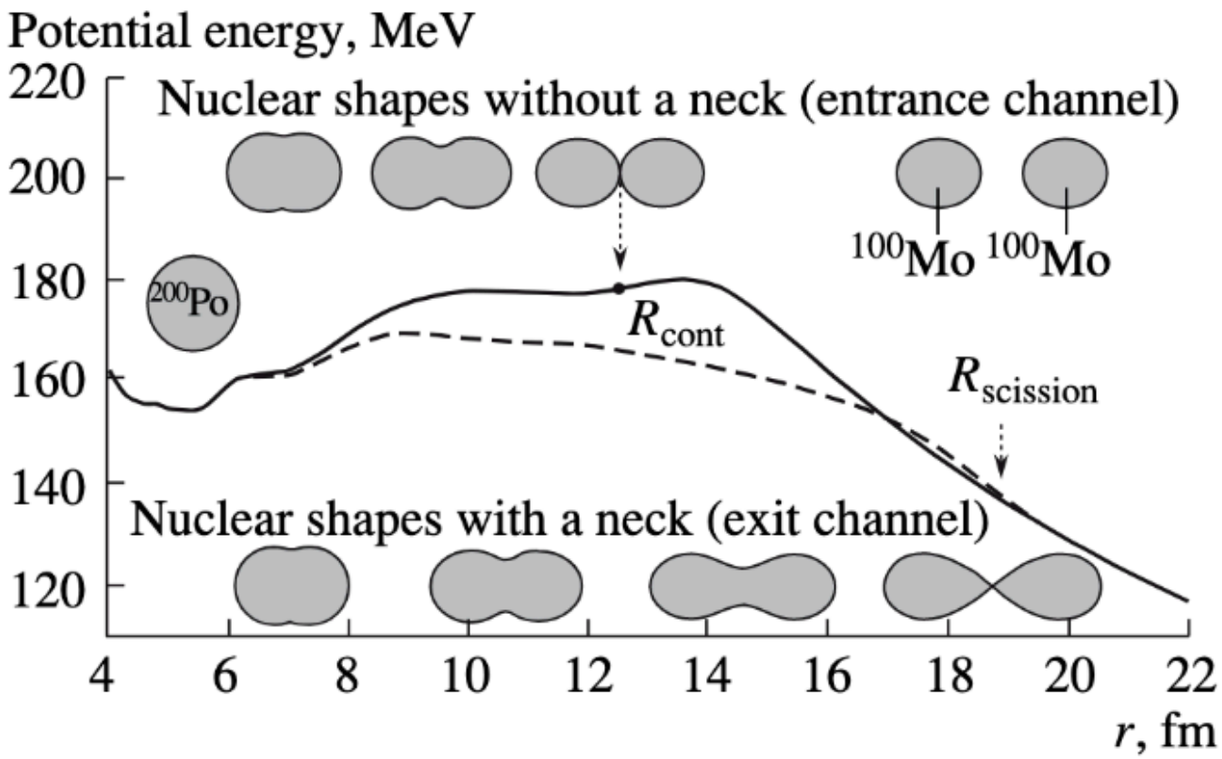
Potential energy surface - Adiabatic potential energy

The time evolution of the neck degree of freedom ε is considered in the same way as the transition from the diabatic to the adiabatic regime.

$$V(q,t) = V_{\text{adiab}}(q, \varepsilon = 1.0) f(t) + V_{\text{adiab}}(q, \varepsilon = 0.35)[1 - f(t)]$$

$$f(t) = \exp(-\frac{t}{\tau_\varepsilon}), \quad \tau_\varepsilon = 10^{-22}\text{s}$$

The entrance and the exit channel potentials are calculated with ε = 1 (no neck) and with ε = 0.35, respectively.



Equations of motion and nucleon transfer

$$W^{\pm} = \lambda_0 \sqrt{\frac{\rho(A \pm 1)}{\rho(A)}} P_{tr}(R, \beta, A \rightarrow A \pm 1)$$

Diffusion constant

It can be qualitatively assumed to be related to the size of the system and the temperature.

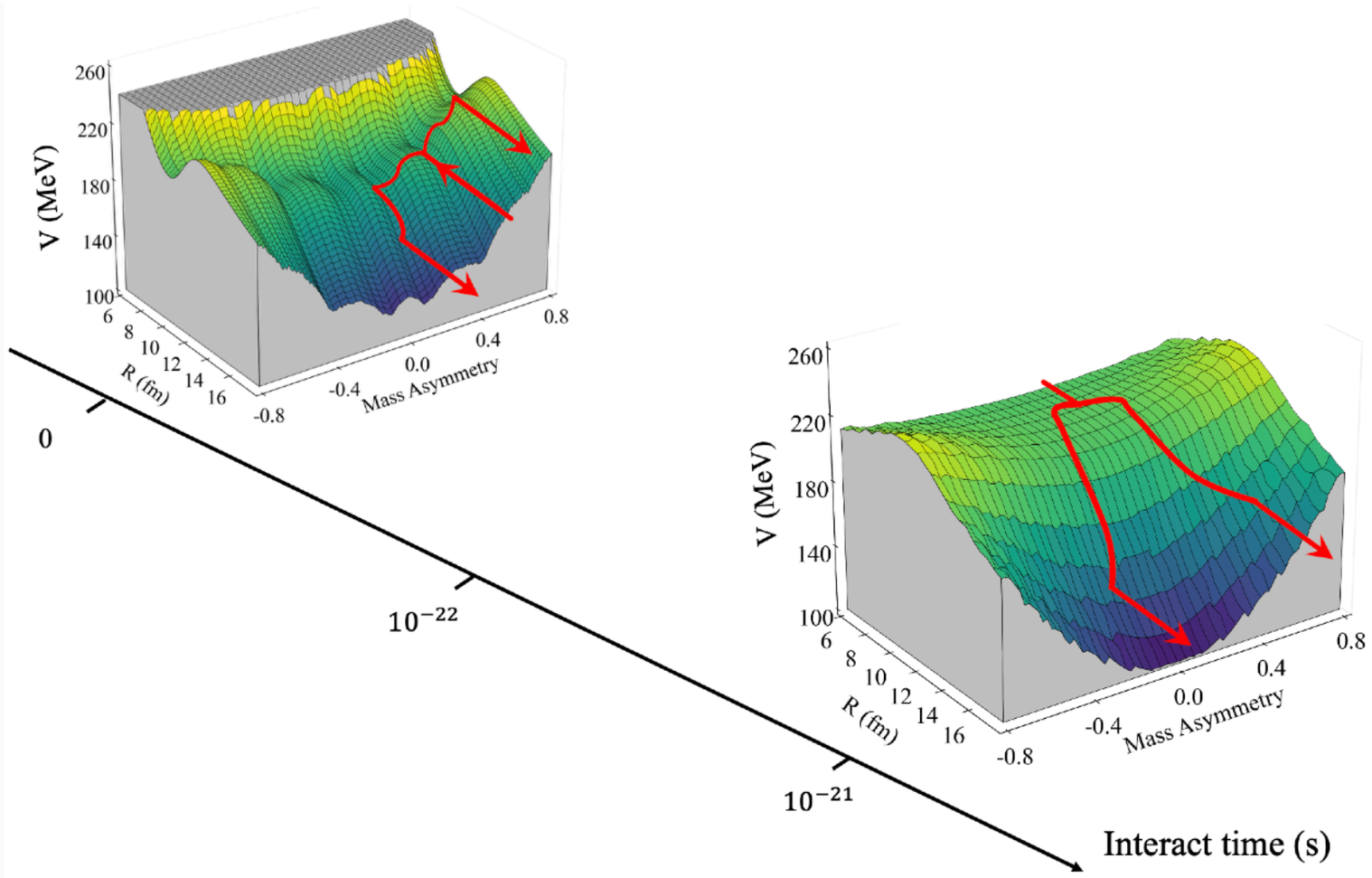
Therefore, we treat it as a free parameter.

	Value	Ref
Moretto et. al.	$2\pi k \frac{R_1 R_2}{R_1 + R_2}, \quad k = 10^{21}(\text{unit/fm/s})$	**Phys. Lett. B (1975) 58 26**
Zagrebaev et. al.	$0.1 * 10^{22}(\text{unit/s})$	**Nucl. Phys. A (2015) 944, 257.**
Karpov et. al.	$5 * A_{tot} * (T/\text{MeV}) * 10^{16}(\text{unit/s})$	**Eur. Phys. J. A (2022) 58:41**

This value is adopted in the model

V. Zagrebaev, A. Karpov, Y. Aritomo, M. Naumenko, and W. Greiner, Phys. Part. Nucl. 38, 469 (2007).

Potential energy surface



During the nuclear reaction process, we also considered the temperature effect

$$E_{\mathrm{mic}}(q,T) = E_{\mathrm{mic}}(q,T=0) \times \exp(\frac{-aT^2}{E_{\mathrm{d}}}), \quad E_{\mathrm{d}} = 20\mathrm{MeV}$$

V. Zagrebaev, A. Karpov, Y. Aritomo, M. Naumenko, and W. Greiner, Phys. Part. Nucl. 38, 469 (2007).

Potential energy surface - Diabatic potential energy

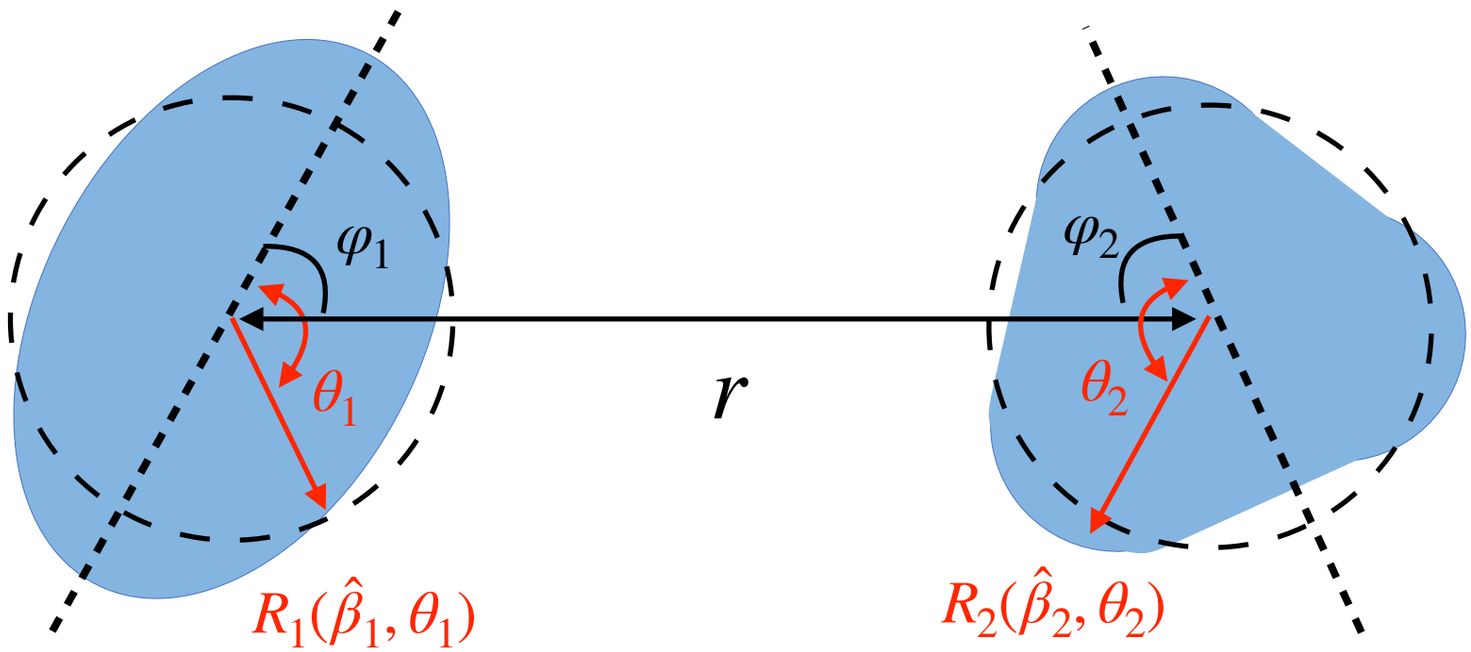
Redistribution of nucleons between separated nuclei changes their binding energies. The diabatic potential energy is defined as follows:

$$V_{\rm diab}(A,Z;q) = V_{12}(A,Z;q) + M(A_1,Z_1;\delta_1) + M(A_2,Z_2;\delta_2) - M(A_T,Z_T;\delta_T^{\rm g.s.}) - M(A_P,Z_P;\delta_P^{\rm g.s.})\,.$$

the masses of the reaction fragments

The interaction energy of the nuclei

the masses of the entrance channel



Potential energy surface - Diabatic potential energy

We apply a phenomenon procedure to its calculation:

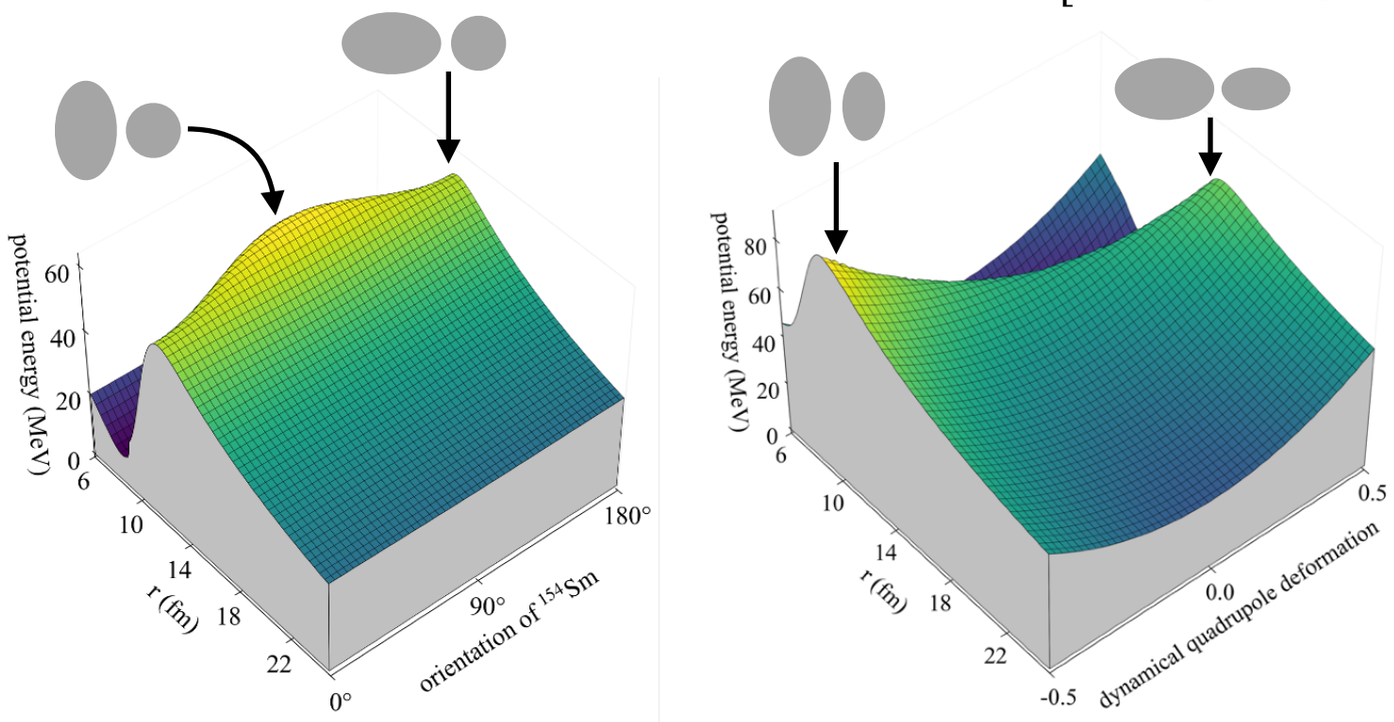
$$V_{12}(A,Z;q) = V_{12}^{C}(A,Z;q) + V_{12}^{N}(A,Z;q) + V^{\mathrm{cent}}(A,Z;r)$$

Coulomb potential:

$$V_{12}^{C}(A,Z;q) \approx V_{C}(A,Z;r,\beta_P^0,\beta_T^0,\theta_P,\theta_T) = \frac{Z_1Z_2e^2}{R} + \left(\frac{9}{20\pi}\right)^{\frac{1}{2}}\left(\frac{Z_1Z_2e^2}{R^3}\right) \times \sum_{i=P,T} R_{0i}^2\beta_i^0 P_2(\cos\theta_i) + \left(\frac{3}{7\pi}\right)\left(\frac{Z_1Z_2e^2}{R^3}\right) \times \sum_{i=P,T} R_{0i}^2[\beta_i^0 P_2(\cos\theta_i)]^2,$$

Nuclear potential:

$$V_{12}^{N}(A,Z;q) \approx V_{N}(A,Z;r,\beta_p^0,\beta_T^0,\theta_p,\theta_T) = \frac{-V_0}{1+\exp\left[(r-R_P-R_T)/a\right]}$$



In the future, I would like to use double folding procedure